

AREA UNDER CURVE

JEE EXERCISES

Exercise J-I

AREA BOUNDED BETWEEN TWO CURVES

1. The area between the curve $xy = a^2$, x-axis, $x = a$ and $x = 2a$ is :
(A) $a \log 2$ (B) $a^2 \log 2$ (C) $2a \log 2$ (D) None of these

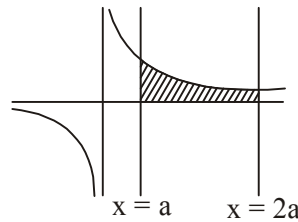
वक्र $xy = a^2$, x-अक्ष, $x = a$ तथा $x = 2a$ के मध्य क्षेत्रफल है :

- (A) $a \log 2$ (B) $a^2 \log 2$ (C) $2a \log 2$ (D) कोई नहीं

Ans. B

Sol. $xy = a^2$ x-axis $x = a$ and $x = 2a$

$$\begin{aligned} A &= \int_a^{2a} y dx \\ &= \int_a^{2a} \frac{a^2}{x} dx = a^2 \log x \Big|_a^{2a} \\ &= a^2 \log \frac{2a}{a} \\ &= a^2 \log 2 \end{aligned}$$



2. Area under the curve $y = \sin 2x + \cos 2x$ between $x = 0$ and $x = \frac{\pi}{4}$, is :

- (A) 2 sq. units (B) 1 sq. units (C) 3 sq. units (D) 4 sq. units

वक्र $y = \sin 2x + \cos 2x$ तथा $x = 0$ तथा $x = \frac{\pi}{4}$, के मध्य क्षेत्रफल है :

- (A) 2 वर्ग इकाई (B) 1 वर्ग इकाई (C) 3 वर्ग इकाई (D) 4 वर्ग इकाई

Ans. B

Sol. $y = \sin 2x + \cos 2x$ $x = 0$ & $\frac{\pi}{4}$

$$\begin{aligned} A &= \int_0^{\frac{\pi}{4}} (\sin 2x + \cos 2x) dx \\ &= -\frac{\cos 2x}{2} + \sin \frac{2x}{2} \Big|_0^{\frac{\pi}{4}} \\ &= \frac{1}{2} + \frac{1}{2} = 1 \end{aligned}$$

3. The area bounded between the curve $|y| = 1 - x^2$ is :
 (A) $2/3$ (B) $4/3$ (C) $8/3$ (D) None of these

वक्र $|y| = 1 - x^2$ के द्वारा परिबद्ध क्षेत्रफल है :

- (A) $2/3$ (B) $4/3$ (C) $8/3$ (D) इनमें से कोई नहीं

Ans. C

Sol. $|y| = 1 - x^2$

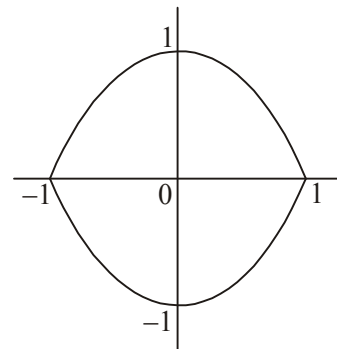
$$y = 1 - x^2 \Rightarrow x^2 = -(y - 1)$$

$$|y| = 1 - x^2$$

$$A = 2 \int_{-1}^1 y dx$$

$$= 2 \int_{-1}^1 (1 - x^2) dx$$

$$= 2 \left(x - \frac{x^3}{3} \right)_{-1}^1 = \left(2 - \frac{1}{3} - \left(-1 + \frac{1}{3} \right) \right) = \frac{8}{3}$$



4. The area bounded by the parabola $y^2 = 4ax$, x-axis and $x = 2a$ and latus rectum is :

परवलय $y^2 = 4ax$, x-अक्ष तथा $x = 2a$ तथा नाभि लम्ब के मध्य क्षेत्रफल है :

- (A) $2a^2 (\sqrt{2} - 1)$ (B) $4a^2 (2\sqrt{2} - 1)$ (C) $\frac{4}{3}a^2 (2\sqrt{2} - 1)$ (D) $\frac{8a^2}{3} (2\sqrt{2} - 1)$

Ans. C

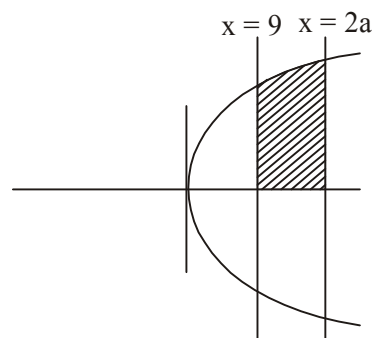
Sol. $y^2 = 4ax$ x-axis $x = 2a$ & latus rectum

$$y = \int_a^{2a} \sqrt{4ax} dx$$

$$= 2\sqrt{a} \left[\frac{2}{3} x^{\frac{3}{2}} \right]_a^{2a}$$

$$= 2\sqrt{a} \left[\frac{2}{3} \cdot 2\sqrt{2}a^{\frac{3}{2}} - \frac{2}{3}a^{\frac{3}{2}} \right]$$

$$= \frac{4}{3} [2\sqrt{2}a^2 - a^2]$$



$$= \frac{4}{3} a^2 [2\sqrt{2} - 1]$$

5. The area bounded between parabola $x^2 = 4y$ and $y = |x|$ is :

परवलय $x^2 = 4y$ तथा $y = |x|$ के मध्य क्षेत्रफल है :

- (A) $2/3$ (B) $4/3$ (C) $8/3$ (D) $16/3$

Ans. D

Sol. $x^2 = 4y$

$$\text{Area} = 2 \int_0^4 (y_1 - y_2) dx$$

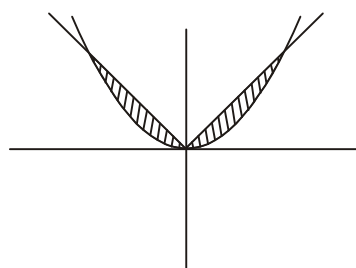
$$= 2 \int_0^4 \left(x - \frac{x^2}{4} \right) dx$$

$$= 2 \left(\frac{x^2}{2} - \frac{x^3}{12} \right) \Big|_0^4$$

$$= 2 \left[\left(8 - \frac{16}{3} \right) - 0 \right]$$

$$= 2 \times \frac{8}{3}$$

$$= \frac{16}{3}$$



6. The area of the figure bounded by the parabola $y = x^2 + 1$ and the straight line $x + y = 3$ is :

$y = x^2 + 1$ तथा सरल रेखा $x + y = 3$ के मध्य क्षेत्रफल है-

- (A) $3/2$ (B) $5/2$ (C) $7/2$ (D) $9/2$

Ans. D

Sol. $y = x^2 + 1$

$x + y = 3$

Point of Intersection

$$3 - x = x^2 + 1$$

$$x^2 + x - 2 = 0$$

$$(x + 2)(x - 1) = 0$$

$$x = -2, 1$$

$$A = \int_{-2}^1 (y_1 - y_2) dx$$

$$= \int_{-2}^1 ((3-x) - (x^2 + 1)) dx$$

$$= \int_{-2}^1 (2 - x - x^2) dx$$

$$= 2x - \frac{x^2}{2} - \frac{x^3}{3} \Big|_{-2}^1$$

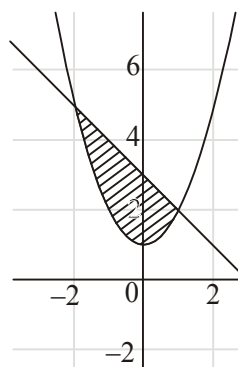
$$A = \left(2 - \frac{1}{2} - \frac{1}{3} \right) - \left(-4 - \frac{4}{2} + \frac{8}{3} \right)$$

$$= \frac{12-3-2}{6} - \frac{-24-12+16}{6}$$

$$= \frac{7}{6} + \frac{20}{6}$$

$$= \frac{27}{6}$$

$$= \frac{9}{2}$$



7. Common area between the parabolas $y = 2x^2$ and $y = x^2 + 4$ is :

- (A) $16/3$ (B) $8/3$ (C) $32/3$ (D) None of these

परवलयों $y = 2x^2$ तथा $y = x^2 + 4$ का उभयनिष्ठ क्षेत्रफल है :

- (A) $16/3$ (B) $8/3$ (C) $32/3$ (D) इनमें से कोई नहीं

Ans. C

Sol. $y = 2x^2$ $y = x^2 + 4$

Point of intersection

$$2x^2 = x^2 + 4$$

$$x^2 = 4$$

$$x = \pm 2$$

$$\text{Area} = 2 \int_0^2 (y_2 - y_1) dx$$

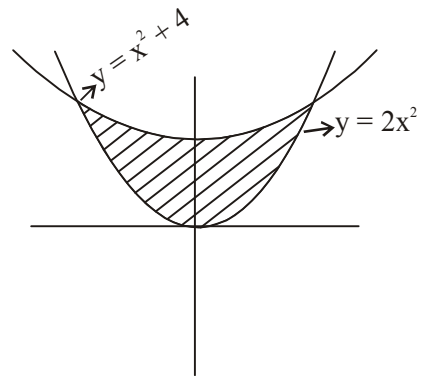
$$A = 2 \int_0^2 ((x^2 + 4) - (2x^2)) dx$$

$$= 2 \int_0^2 (4 - x^2) dx$$

$$= 2 \left[4x - \frac{x^3}{3} \right]_0^2$$

$$= 2 \left(8 - \frac{8}{3} \right)$$

$$= \frac{32}{3}$$



8. Area bounded by the curves $y = 2^x$, $y = 2x - x^2$, $x = 0$ and $x = 2$ is :

वक्रों $y = 2^x$, $y = 2x - x^2$, $x = 0$ तथा $x = 2$ के मध्य क्षेत्रफल है :

- (A) $\frac{3}{\log 2} - \frac{4}{3}$ (B) $\frac{3}{\log 2} + \frac{4}{3}$ (C) $3 \log 2 + \frac{4}{3}$ (D) $3 \log 2 - \frac{4}{3}$

Ans. A

Sol. $y = 2^x$ $y = 2x - x^2$ $x = 0$ $x = 2$

$$y = 2^x$$

$$y = 2x - x^2$$

$$-y = x^2 - 2x$$

$$-(y - 1) = (x - 1)^2$$

$$\text{Area} = \int_0^2 (y_2 - y_1) dx$$

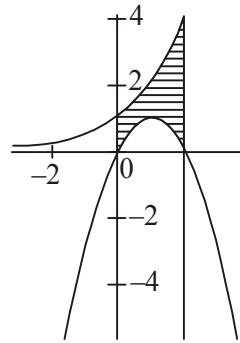
$$= \int_0^2 (2^x - (2x - x^2)) dx$$

$$= \int_0^2 (2^x - 2x + x^2) dx$$

$$= \left. \frac{2^x}{\ln 2} - x^2 + \frac{x^3}{3} \right|_0^2$$

$$= \frac{4}{\ln 2} - 4 + \frac{8}{3} - \frac{1}{\ln 2}$$

$$= \frac{3}{\ln 2} - \frac{4}{3}$$



9. The area of the closed figure bounded by $y = \cos x$, $y = 1 + (2/\pi)x$ and $x = \pi/2$ is :

$y = \cos x$, $y = 1 + (2/\pi)x$ तथा $x = \pi/2$ के मध्य क्षेत्रफल है :

(A) $\frac{3\pi}{4}$

(B) $\frac{\pi}{2}$

(C) $\frac{3\pi - 4}{4}$

(D) $\frac{\pi - 4}{4}$

Ans. C

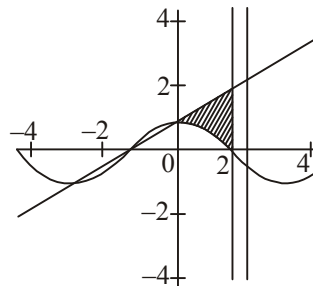
Sol. $y = \cos x$ $y = 1 + \frac{2}{\pi}x$ $x = \frac{\pi}{2}$

$$A = \int_0^{\frac{\pi}{2}} (y_2 - y_1) dx$$

$$= \int_0^{\frac{\pi}{2}} \left(\left(1 + \frac{2}{\pi}x \right) - \cos x \right) dx$$

$$= \left. x + \frac{x^2}{\pi} - \sin x \right|_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2} + \frac{\pi}{4} - 1$$



$$= \frac{3\pi}{4} - 1$$

10. The area enclosed between the curves $y = \log_e (x + e)$, $x = \log_e \left(\frac{1}{y} \right)$ and the x-axis is :

चक्रों $y = \log_e (x + e)$, $x = \log_e \left(\frac{1}{y} \right)$ तथा x-अक्ष द्वारा बने क्षेत्र का क्षेत्रफल होगा :

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $y = \log_e (x + e)$ $x = \log_e \left(\frac{1}{y} \right)$ x-axis

$$x = \log_e \left(\frac{1}{y} \right)$$

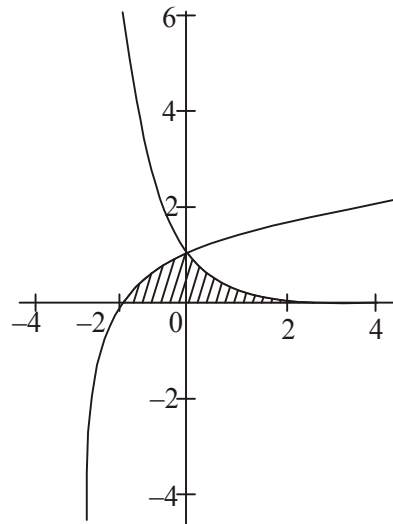
$$A = \int_{1-e}^0 \log_e (x + e) dx - \int_0^{\infty} e^{-x} dx$$

$$= (x + e) \log_e \left(\frac{x + e}{e} \right) \Big|_{1-e}^0 - e^{-x} \Big|_0^{\infty}$$

$$= \left(-\log_e \frac{1}{e} \right) - (-1)$$

$$= 1 + 1$$

$$= 2$$



11. Area of the circle $(x - 2)^2 + (y - 3)^2 = 32$ which lies below the line $y = x + 1$ is :

- (A) $\int_{-2}^6 \left[(x+1) + \sqrt{32 - (x-2)^2} + 3 \right] dx$ (B) $\int_{-2}^6 \left[\left\{ \sqrt{32 - (x-2)^2} + 3 \right\} - (x+1) \right] dx$
 (C) 16π (D) None of these

वृत्त $(x - 2)^2 + (y - 3)^2 = 32$ जो कि रेखा $y = x + 1$ के नीचे स्थित है, का क्षेत्रफल है :

- (A) $\int_{-2}^6 \left[(x+1) + \sqrt{32 - (x-2)^2} + 3 \right] dx$ (B) $\int_{-2}^6 \left[\left\{ \sqrt{32 - (x-2)^2} + 3 \right\} - (x+1) \right] dx$
 (C) 16π (D) इनमें से कोई नहीं

Ans. C

Sol. $(x-2)^2 + (y-3)^2 = 32$ $y = x + 1$

Point of intersection

$$(x-2)^2 + (x-2)^2 = 32$$

$$(x-2)^2 = 16$$

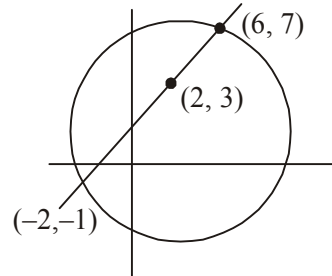
$$x-2 = \pm 4$$

$$x = 6, -2$$

$$x = 6 \quad y = 7 \quad (6, 7)$$

$$x = -2 \quad y = -1 \quad (-2, -1)$$

center is on the line, so line is diameter



$$A = \frac{\pi r^2}{2} = \frac{\pi \times 32}{2} = 16\pi$$

12. If $y = 2 \sin x + \sin 2x$ for $0 \leq x \leq 2\pi$, then the area enclosed by the curve and the x-axis is :

- (A) 6 (B) 8 (C) 12 (D) none

यदि $y = 2 \sin x + \sin 2x$, $0 \leq x \leq 2\pi$ के लिये, तो वक्र तथा x-अक्ष द्वारा परिबद्ध क्षेत्रफल होगा :

- (A) 6 (B) 8 (C) 12 (D) कोई नहीं

Ans. B

Sol. $y = 2 \sin x + \sin 2x$ x-axis $x \in [0, 2\pi]$

$$y = 2 \sin x (1 + \cos x)$$

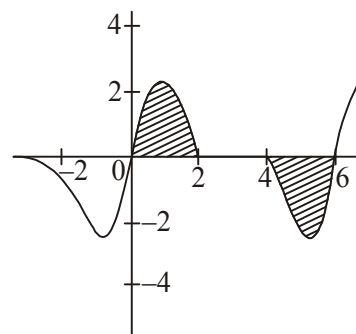
$$= 2 \sin x \cdot 2 \cos^2 \frac{x}{2}$$

$$A = \int_0^\pi (2 \sin x + \sin 2x) dx + (-) \int_\pi^{2\pi} 2 \sin x + \sin 2x$$

$$= -2 \cos x - \frac{\cos 2x}{2} \Big|_0^\pi - \left[-2 \cos x - \frac{\cos 2x}{2} \right]_\pi^{2\pi}$$

$$= +2 - \frac{1}{2} + 2 + \frac{1}{2} - \left[-2 - \frac{1}{2} - \left(2 - \frac{1}{2} \right) \right]$$

$$= 8$$



13. The area of the closed figure bounded by $x = -1$, $x = 2$ and $y = \begin{cases} -x^2 + 2, & x \leq 1 \\ 2x - 1, & x > 1 \end{cases}$ and the abscissa axis, is :

वक्र $x = -1$, $x = 2$, $y = \begin{cases} -x^2 + 2, & x \leq 1 \\ 2x - 1, & x > 1 \end{cases}$ तथा x -अक्ष द्वारा परिबद्ध क्षेत्रफल होगा :

- (A) $\frac{7}{3}$ (B) $\frac{10}{3}$ (C) $\frac{13}{3}$ (D) $\frac{16}{3}$

Ans. D

Sol. $y = \begin{cases} -x^2 + 2 & x \leq 1 \\ 2x - 1 & x > 1 \end{cases}$, $x = -1, x = 2$

$$A = \int_{-1}^1 (-x^2 + 2) dx + \int_1^2 (2x - 1) dx$$

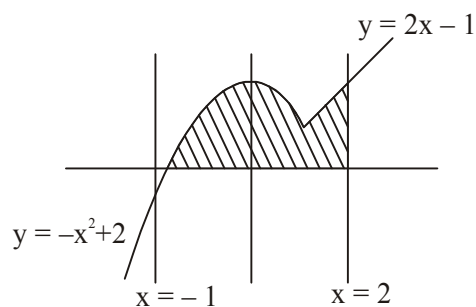
$$= 2 \int_0^1 (-x^2 + 2) dx + x^2 - x \Big|_1^2$$

$$= 2 \left(\frac{-x^3}{3} + 2x \right) \Big|_0^1 + ((4 - 2) - (1 - 1))$$

$$= 2 \left(-\frac{1}{3} + 2 \right) + 2$$

$$= \frac{10}{3} + 2$$

$$= \frac{16}{3}$$



14. Area between curve $y = \frac{x}{1+x^2}$ and x axis from $x = -1$ to $x = 1$ is :

वक्र $y = \frac{x}{1+x^2}$ तथा x -अक्ष, $x = -1$ से $x = 1$ तक परिबद्ध क्षेत्रफल होगा :

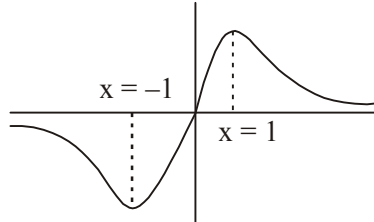
- (A) $\ln 2$ (B) $\ln 4$ (C) $\pi/2$ (D) π

Ans. A

Sol. $y = \frac{x}{1+x^2}$ x-axis $x = -1, 1$

$$\frac{dy}{dx} = \frac{1+x^2-2x^2}{(1+x^2)^2}$$

$$= \frac{1-x^2}{(1+x^2)^2}$$



$$A = 2 \int_0^1 y dx$$

$$= 2 \int_0^1 \frac{x}{1+x^2} dx$$

$$= \log(1+x^2) \Big|_0^1$$

$$= \log 2$$

15. The area of the figure bounded by the curves $y = e^x$, $y = e^{-x}$ and the straight line $x = 1$ is :

- (A) $e + \frac{1}{e}$ (B) $e - \frac{1}{e}$ (C) $e + \frac{1}{e} - 2$ (D) none

वक्र $y = e^x$, $y = e^{-x}$ तथा सरल रेखा $x = 1$ द्वारा परिबद्ध क्षेत्रफल होगा :

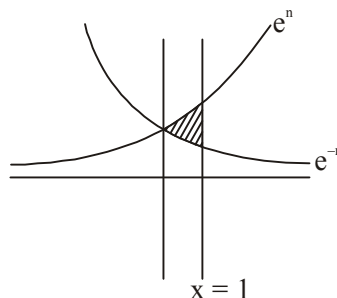
- (A) $e + \frac{1}{e}$ (B) $e - \frac{1}{e}$ (C) $e + \frac{1}{e} - 2$ (D) कोई नहीं

Ans. C

Sol. $y = e^x$ $y = e^{-x}$ $x = 1$

$$A = \int_0^1 e^x dx - \int_0^1 e^{-x} dx$$

$$= (e^x)_0^1 - (-e^{-x})_0^1$$



$$= (e-1) - \left(-\frac{1}{e} + 1\right)$$

$$= e + \frac{1}{e} - 2$$

16. The area bounded by $y = x^2$ and $y = 2 - x^2$ is :

वक्र $y = x^2$ तथा $y = 2 - x^2$ द्वारा परिबद्ध क्षेत्रफल होगा :

- (A) $\frac{8}{3}$ (B) $\frac{16}{3}$ (C) $\frac{32}{3}$ (D) $\frac{64}{3}$

Ans. A

Sol. $y = x^2$ $y = 2 - x^2$

Point of intersection $x = \pm 1$

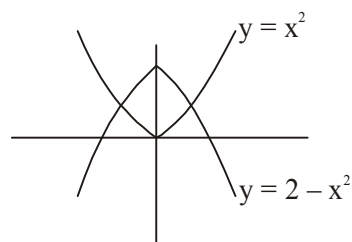
$$A = 2 \int_0^1 (y_2 - y_1) dx$$

$$= 2 \int_0^1 (2 - x^2 - x^2) dx$$

$$= 2 \left(2x - \frac{2x^3}{3} \right)_0^1$$

$$= 2 \left(2 - \frac{2}{3} \right)$$

$$= \frac{8}{3}$$



17. The area bounded by the curves $y^2 = 3x$ & $y^2 = 36 - 9x$ is :

- (A) 16 (B) 18 (C) 21.5 (D) none

वक्र $y^2 = 3x$ तथा $y^2 = 36 - 9x$ द्वारा परिबद्ध क्षेत्रफल होगा :

- (A) 16 (B) 18 (C) 21.5 (D) कोई नहीं

Ans. A

Sol. $y^2 = 3x$ $y^2 = 36 - 9x$

$$= -9(x-4)$$

Point of intersection

$$3x = 36 - 9x$$

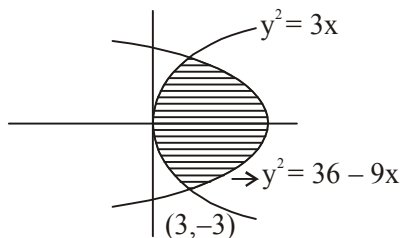
$$12x = 36$$

$$x = 3$$

$$y^2 = 9$$

$$y = \pm 3$$

$$(3, 3) \quad (3, -3)$$



$$A = 2 \int_0^3 (x_2 - x_1) dy$$

$$= 2 \int_0^3 \left(\frac{36 - y^2}{9} - \left(\frac{y^2}{3} \right) \right) dy$$

$$= 2 \left[\frac{36y - \frac{y^3}{3}}{9} - \frac{y^3}{9} \right]_0^3$$

$$= 2 \left[\frac{108 - 9}{9} - 3 \right]$$

$$= 2 \left[\frac{99}{9} - 3 \right]$$

$$= 2 [11 - 3]$$

$$= 16$$

18. The ratio in which the area enclosed by the curve $y = \cos x$ $\left(0 \leq x \leq \frac{\pi}{2} \right)$ in the first quadrant is divided by the curve $y = \sin x$, is :

(A) $(\sqrt{2} - 1) : 1$ (B) $(\sqrt{2} + 1) : 1$ (C) $\sqrt{2} : 1$ (D) none

vuŋk gkɛk:

- Ans.** C

$$\frac{A_1}{A_2} = ?$$

$$= \sqrt{2} - 1$$

$$= -\cos x \Big|_0^{\frac{\pi}{4}} + \sin x \Big|_{\frac{\pi}{2}}^{\frac{\pi}{4}}$$

$$= -\frac{1}{\sqrt{2}} + 1 + 1 - \frac{1}{\sqrt{2}}$$

$$= 2 - \sqrt{2}$$

$$= \sqrt{2}(\sqrt{2}-1)$$

$$\frac{A_2}{A_1} = \frac{\sqrt{2}(\sqrt{2}-1)}{\sqrt{2}-1} = \frac{\sqrt{2}}{1}$$

19. The area bounded by the curve $y^2 + x^4 = x^2$ is :

- (A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) $\frac{8}{3}$ (D) none

वक्र $y^2 + x^4 = x^2$ से घिरा हुआ क्षेत्रफल होगा :

- (A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) $\frac{8}{3}$ (D) कोई नहीं

Ans. B

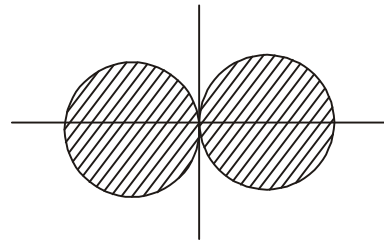
Sol. $y^2 + x^4 = x^2$

$$y^2 = x^2 - x^4$$

$$y^2 = x^2 (1 - x^2)$$

$$y = \pm x \sqrt{1 - x^2}$$

Symmetric about x-axis and y-axis both



$$A = 4 \int_0^1 y dx$$

$$= 4 \int_0^1 x \sqrt{1 - x^2} dx$$

$$= 2 \left(-\frac{2}{3} (1 - x^2)^{\frac{3}{2}} \right) \Big|_0^1$$

$$= 2 \left(\frac{2}{3} \right) = \frac{4}{3}$$

20. Area bounded by the curves $y = |x| - 2$ and $y = 1 - |x - 1|$ is equal to :

- (A) 4 sq. units (B) 6 sq. units (C) 2 sq. units (D) 8 sq. units

वक्रों $y = |x| - 2$ तथा $y = 1 - |x - 1|$ के मध्य घिरा हुआ क्षेत्रफल होगा :

- (A) 4 वर्ग इकाई (B) 6 वर्ग इकाई (C) 2 वर्ग इकाई (D) 8 वर्ग इकाई

Ans. A

Sol. $y = |x| - 2$ $y = 1 - |x - 1|$

$$I \quad y = |x| - 2$$

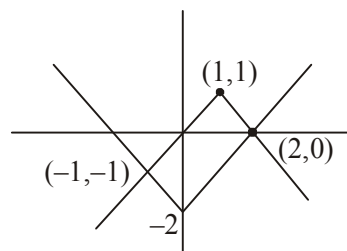
$$y = \begin{cases} x-2 & x \geq 0 \\ -x-2 & x < 0 \end{cases}$$

II $y = 1 - |x - 1|$

$$y = \begin{cases} 2-x & x \geq 1 \\ x & x < 1 \end{cases}$$

$$A = l \times b$$

$$= 2\sqrt{2} \times \sqrt{2} = 4$$



21. Area bounded by the parabola $y = x^2 - 2x + 3$ and tangents drawn to it from the point $P(1, 0)$ is equal to :

- (A) $4\sqrt{2}$ sq. units (B) $\frac{4\sqrt{2}}{3}$ sq. units (C) $\frac{8\sqrt{2}}{3}$ sq. units (D) $\frac{16}{3}\sqrt{2}$ sq. units

परवलय $y = x^2 - 2x + 3$ तथा बिन्दु $P(1, 0)$ से इस पर खींची गई स्पर्श रेखा के मध्य का क्षेत्रफल होगा :

- (A) $4\sqrt{2}$ वर्ग ईकाई (B) $\frac{4\sqrt{2}}{3}$ वर्ग ईकाई (C) $\frac{8\sqrt{2}}{3}$ वर्ग ईकाई (D) $\frac{16}{3}\sqrt{2}$ वर्ग ईकाई

Ans. B

Sol. $y = x^2 - 2x + 3$ tangent from $P(1, 0)$

$$y - 2 = (x - 1)^2$$

Tangent

$$x = \frac{y-2}{m} + \frac{m}{4}$$

$$(1, 0)$$

$$= \frac{-2}{m} + \frac{m}{4}$$

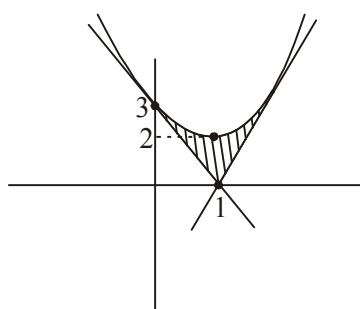
$$0 = -8 + m^2$$

$$m = \pm 2\sqrt{2}$$

Tangents

$$y = 2\sqrt{2}(x-1) \quad \& \quad y = -2\sqrt{2}(x-1)$$

$$\text{Point of intersection of } y - 2 = (x - 1)^2 \quad \& \quad y = \pm 2\sqrt{2}(x - 1)$$



$$y - 2 = \frac{y^2}{8} \Rightarrow y^2 - 8y + 16 = 0$$

$$y = 4$$

$$A = 2 \int_1^{1+\sqrt{2}} (y_1 - y_2) dx$$

$$= 2 \int_1^{1+\sqrt{2}} \left[2 + (x-1)^2 - 2\sqrt{2(x-1)} \right] dx$$

$$x - 1 = t$$

$$= 2 \int_0^{\sqrt{2}} (2 + t^2 - 2\sqrt{2}t) dt$$

$$= 2 \left(2t + \frac{t^3}{3} - \sqrt{2}t^2 \right)_0^{\sqrt{2}}$$

$$= 2 \left(2\sqrt{2} + \frac{2\sqrt{2}}{3} - 2\sqrt{2} \right)$$

$$= \frac{4\sqrt{2}}{3}$$

22. Area bounded by the curves $y = \sin x$, tangent drawn to it at $x = 0$ and the line $x = \frac{\pi}{2}$, is equal to :

- (A) $\frac{\pi^2 - 4}{2}$ sq. units (B) $\frac{\pi^2 - 8}{8}$ sq. units (C) $\frac{\pi^2 - 2}{4}$ sq. units (D) $\frac{\pi^2 - 2}{2}$ sq. units

वक्रों $y = \sin x$, तथा इस पर $x = 0$ पर स्पर्श रेखा तथा रेखा $x = \frac{\pi}{2}$ के मध्य का क्षेत्रफल होगा :

- (A) $\frac{\pi^2 - 4}{2}$ वर्ग ईकाई (B) $\frac{\pi^2 - 8}{8}$ वर्ग ईकाई (C) $\frac{\pi^2 - 2}{4}$ वर्ग ईकाई (D) $\frac{\pi^2 - 2}{2}$ वर्ग ईकाई

Ans. B

Sol. $y = \sin x$ Tangent at $x = 0$ & line $x = \frac{\pi}{2}$

$$\frac{dy}{dx} = \cos x$$

$$m_T(x=0) = 1$$

$$\text{at } x=0 \quad y=0$$

$$\text{Tangent } y=mx$$

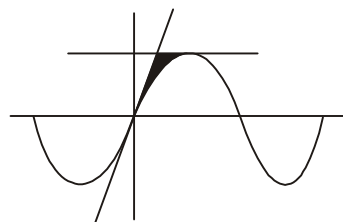
$$y=x$$

$$A = \int_0^{\frac{\pi}{2}} (-\sin x + x) dx$$

$$= \left(+\cos x + \frac{x^2}{2} \right) \Big|_0^{\frac{\pi}{2}}$$

$$= \left(+\cos \frac{\pi}{2} + \frac{\pi^2}{8} \right) - (1)$$

$$= +\frac{\pi^2}{8} - 1 = \frac{\pi^2 - 8}{8}$$



23. Value of the parameter a such that the area bounded by $y = a^2 x^2 + ax + 1$, coordinate axes and the line $x = 1$, attains its least value, is equal to :

प्राचल 'a' के किस मान के लिए वक्र $y = a^2 x^2 + ax + 1$, निर्देशांक अक्षों को $x = 1$ से घिरे हुये क्षेत्रफल का न्यूनतम मान होगा :

- (A) $-\frac{1}{4}$ (B) $-\frac{1}{2}$ (C) $-\frac{3}{4}$ (D) -1

Ans. C

Sol. $y = a^2 x^2 + ax + 1$ x-axis & y-axis line $x = 1$

$$D = a^2 - 4(a^2)$$

$$= -3a^2 < 0$$

$$A = \int_0^1 y dx$$

$$A = \int_0^1 (a^2 x^2 + ax + 1) dx$$

$$A = \frac{a^2 x^3}{3} + \frac{ax^2}{2} + x \Big|_0^1$$

$$A = \frac{a^2}{3} + \frac{a}{2} + 1$$

$$\frac{dA}{da} = \frac{2a}{3} + \frac{1}{2} = 0$$

$$a = -\frac{3}{4}$$

$$A = \frac{9}{16 \times 3} - \frac{3}{8} + 1$$

$$= \frac{3}{16} - \frac{6}{16} + \frac{16}{16}$$

$$= \frac{13}{16}$$

24. Area enclosed by the curve $y = f(x)$ defined parametrically as $x = \frac{1-t^2}{1+t^2}$, $y = \frac{2t}{1+t^2}$ is equal to :

- (A) π sq. units (B) $\frac{\pi}{2}$ sq. units (C) $\frac{3\pi}{4}$ sq. units (D) $\frac{3\pi}{2}$ sq. units

यदि वक्र $y = f(x)$ प्राचलिक रूप में $x = \frac{1-t^2}{1+t^2}$, $y = \frac{2t}{1+t^2}$ प्रकार से परिभाषित है, तो इस से घिरा हुआ क्षेत्रफल होगा :

- (A) π वर्ग इकाई (B) $\frac{\pi}{2}$ वर्ग इकाई (C) $\frac{3\pi}{4}$ वर्ग इकाई (D) $\frac{3\pi}{2}$ वर्ग इकाई

Ans. A

Sol. $y = f(x)$

$$x = \frac{1-t^2}{1+t^2} \quad \& \quad y = \frac{2t}{1+t^2}$$

Let $t = \tan \theta$

$$x = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

$$x = \cos 2\theta$$

$$\text{and } y = \frac{2 \tan \theta}{1 + \tan^2 \theta}$$

$$y = \sin 2\theta$$

$$x^2 + y^2 = 1$$

$$\text{Area} = \pi r^2$$

$$= \pi \times 1$$

$$= \pi$$

25. The area of the region bounded by the curves $y = |x - 1|$ and $y = 3 - |x|$ is:

- (A) 6 sq. units (B) 2 sq. units (C) 3 sq. units (D) 4 sq. units

वक्र $y = |x - 1|$ तथा $y = 3 - |x|$ के द्वारा परिबद्ध क्षेत्रफल है :

- (A) 6 वर्ग इकाई (B) 2 वर्ग इकाई (C) 3 वर्ग इकाई (D) 4 वर्ग इकाई

Ans. D

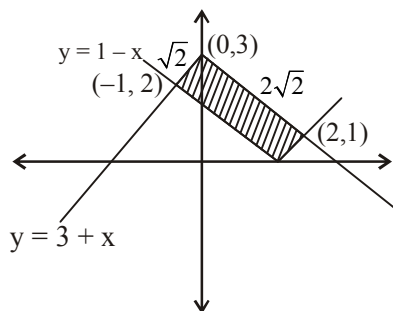
Sol. $y = |x - 1|$ $y = 3 - |x|$

$$y = \begin{cases} x - 1 & x \geq 1 \\ 1 - x & x < 1 \end{cases}$$

$$y = \begin{cases} 3 - x & x > 0 \\ 3 + x & x \leq 0 \end{cases}$$

$$\text{Area} = \sqrt{2} \times 2\sqrt{2}$$

$$= 4$$



26. The area enclosed between the curve $y = \log_e (x + e)$ and the coordinate axes is :

उस क्षेत्रफल, जो कि वक्र $y = \log_e (x + e)$ और अक्षों के बीच में बना है, उसका मान होगा :

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. A

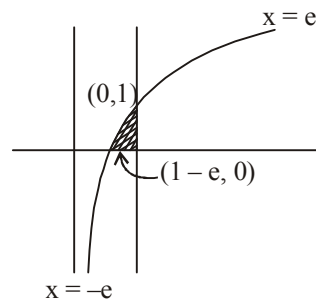
Sol. $y = \log_e (x + e)$

$$A = \int_{1-e}^0 \log_e (x + e) dx$$

$$= (x + e) \log_e \left(\frac{x + e}{e} \right) \Big|_{1-e}^0$$

$$= (1 - e + e) (1)$$

$$= 1$$



27. The parabolas $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the lines $x = 4$, $y = 4$ and the coordinate axes. If S_1 , S_2 , S_3 are respectively the areas of these parts numbered from top to bottom; then $S_1 : S_2 : S_3$ is :

परवलय $y^2 = 4x$ तथा $x^2 = 4y$, रेखाओं $x = 4$, $y = 4$ तथा निर्देशांक अक्षों से घिरे वर्गाकार क्षेत्र को बांटते हैं। यदि ऊपर से नीचे तक क्रमशः S_1 , S_2 तथा S_3 हैं, तो $S_1 : S_2 : S_3$ का मान है :

- (A) 1 : 2 : 1 (B) 1 : 2 : 3 (C) 2 : 1 : 2 (D) 1 : 1 : 1

Ans. D

Sol. $y^2 = 4x$ $x^2 = 4y$

$$S_1 = \int_0^4 x dy$$

$$= \int_0^4 \frac{y^2}{4} dy$$

$$= \frac{y^3}{3 \times 4} \Big|_0^4$$

$$= \frac{64}{3 \times 4} = \frac{16}{3}$$

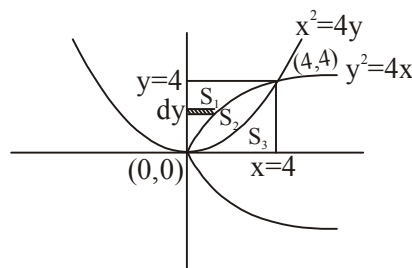
from symmetry

$$S_2 = \frac{16}{3}$$

$$S_2 = 16 - S_1 - S_3$$

$$= 16 - \frac{16}{3} - \frac{16}{3}$$

$$= 16 - \frac{32}{3}$$



$$= \frac{16}{3}$$

$$S_1 : S_2 : S_3 = 1 : 1 : 1$$

28. The area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$ is :

यहाँ $y = |x| - 1$ और $y = -|x| + 1$ से परिबद्ध क्षेत्रफल है :

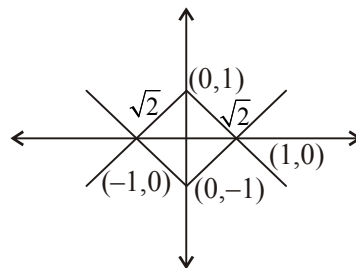
- (A) 1 (B) 2 (C) $2\sqrt{2}$ (D) 4

Ans. B

Sol. $y = |x| - 1$ $y = -|x| + 1$

$$y = \begin{cases} x-1 & x > 0 \\ x-1 & x \leq 0 \end{cases}$$

$$A = \sqrt{2} \times \sqrt{2} = 2$$



29. If the area bounded by $y = ax^2$ and $x = ay^2$, $a > 0$, is 1, then $a =$

यदि $y = ax^2$ और $x = ay^2$, $a > 0$ से परिबद्ध क्षेत्रफल 1 है, तब $a =$

- (A) 1 (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{3}$ (D) $-\frac{1}{\sqrt{3}}$

Ans. B

Sol. $y = ax^2$ $x = ay^2$

$$x^2 = \frac{y}{a} \quad y^2 = \frac{x}{a}$$

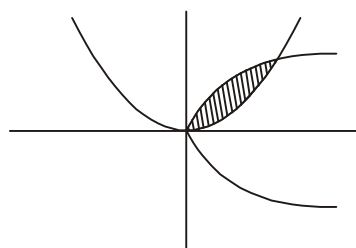
$$\text{Enclosed area} = \frac{16}{3} \left(\frac{1}{4a} \times \frac{1}{4a} \right)$$

$$A = \frac{1}{3a^2}$$

$$1 = \frac{1}{3a^2}$$

$$a = \pm \frac{1}{\sqrt{3}}$$

$$\because a > 0 \quad a = \frac{1}{\sqrt{3}}$$



30. The area bounded by the parabolas $y = (x + 1)^2$ and $y = (x - 1)^2$ and the line $y = 1/4$ is :

- (A) 4 sq. units (B) $\frac{1}{6}$ sq. units (C) $\frac{4}{3}$ sq. units (D) $\frac{1}{3}$ sq. units

परवलयों $y = (x + 1)^2$ और $y = (x - 1)^2$ और रेखा $y = 1/4$ से परिबद्ध क्षेत्रफल है :

- (A) 4 वर्ग इकाई (B) $\frac{1}{6}$ वर्ग इकाई (C) $\frac{4}{3}$ वर्ग इकाई (D) $\frac{1}{3}$ वर्ग इकाई

Ans. D

Sol. $y = (x + 1)^2$

$$y = (x - 1)^2$$

$$y = \frac{1}{4}$$

$$A = \int_{\frac{1}{4}}^1 (-\sqrt{y} + 1) dy$$

$$\because y = (x - 1)^2$$

$$x = \begin{cases} -\sqrt{y} + 1 & \text{for } x < 1 \\ \sqrt{y} + 1 & \text{for } x \geq 1 \end{cases}$$

$$= 2 \int_{\frac{1}{4}}^1 (-\sqrt{y} + 1) dy$$

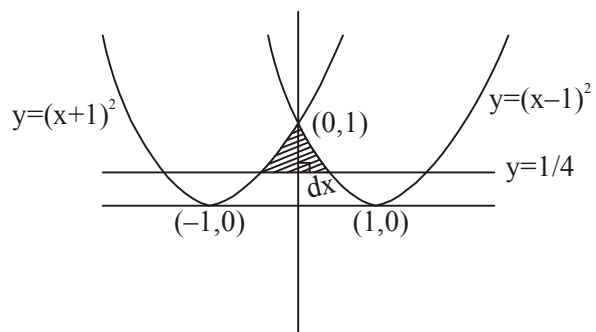
$$= 2 \left(-\frac{2}{3} y^{\frac{3}{2}+1} \right)_{\frac{1}{4}}^1$$

$$= 2 \left[\left(-\frac{2}{3} + 1 \right) - \left(-\frac{2}{3} \times \frac{1}{8} + \frac{1}{4} \right) \right]$$

$$= 2 \left[\frac{1}{3} - \left(-\frac{1}{12} + \frac{1}{4} \right) \right]$$

$$= 2 \left[\frac{1}{3} - \left(\frac{1}{6} \right) \right]$$

$$= 2 \left(\frac{1}{6} \right) = \frac{1}{3}$$



31. The area of the region between the curves $y = \sqrt{\frac{1+\sin x}{\cos x}}$ and $y = \sqrt{\frac{1-\sin x}{\cos x}}$ bounded by the lines $x = 0$ and

$x = \frac{\pi}{4}$ is :

चक्रों $y = \sqrt{\frac{1+\sin x}{\cos x}}$ और $y = \sqrt{\frac{1-\sin x}{\cos x}}$ तथा रेखाओं $x = 0$ और $x = \frac{\pi}{4}$ से परिबद्ध क्षेत्र का क्षेत्रफल है :

(A) $\int_0^{\sqrt{2}-1} \frac{t}{(1+t^2)\sqrt{1-t^2}} dt$

(B) $\int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$

(C) $\int_0^{\sqrt{2}+1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$

(D) $\int_0^{\sqrt{2}+1} \frac{t}{(1+t^2)\sqrt{1-t^2}} dt$

Ans. B

Sol. Area = $\int_0^{\pi/4} \left(\sqrt{\frac{1+\sin x}{\cos x}} - \sqrt{\frac{1-\sin x}{\cos x}} \right) dx$

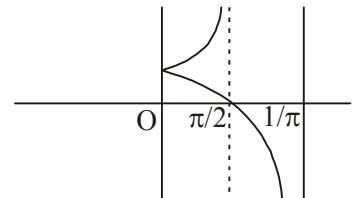
$$= \int_0^{\pi/4} \frac{\left(\cos \frac{x}{2} + \sin \frac{x}{2} \right) - \left(\cos \frac{x}{2} - \sin \frac{x}{2} \right)}{\sqrt{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}} dx$$

$$= \int_0^{\pi/4} \frac{2 \sin \frac{x}{2}}{\sqrt{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}} dx = \int_0^{\pi/4} \frac{2 \tan \frac{x}{2}}{\sqrt{1 - \tan^2 \frac{x}{2}}} dx$$

Let $\tan \frac{x}{2} = t$

$$\sec^2 \frac{x}{2} dx = 2dt \Rightarrow dx = \frac{2dt}{(1+t^2)}$$

$$\therefore \text{Area} = \int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$$



32. The area enclosed between the curves $y^2 = x$ and $y = |x|$ is :

चक्र $y^2 = x$ तथा $y = |x|$ द्वारा परिबद्ध क्षेत्रफल है :

(A) $\frac{2}{3}$

(B) 1

(C) $\frac{1}{6}$

(D) $\frac{1}{3}$

Ans. C

Sol. $y^2 = x$ $y = |x|$

$$A = \int_0^1 (y - y^2) dy$$

$$= \left. \frac{y^2}{2} - \frac{y^3}{3} \right|_0^1$$

$$= \frac{1}{2} - \frac{1}{3}$$

$$= \frac{1}{6}$$

33. The area of the plane region bounded by the curves $x + 2y^2 = 0$ and $x + 3y^2 = 1$ is equal to :

वक्रों $x + 2y^2 = 0$ तथा $x + 3y^2 = 1$ के बीच घिरे तलीय क्षेत्र का क्षेत्रफल है :

(A) $\frac{1}{3}$

(B) $\frac{2}{3}$

(C) $\frac{4}{3}$

(D) $\frac{5}{3}$

Ans. C

Sol. $y^2 = -\frac{x}{2}$ $y^2 = -\frac{1}{3}(x-1)$

$x = -2y^2$ $x = 1 - 3y^2$

POI

$$y^2 = -\frac{x}{2} = -\frac{1}{3}(x-1)$$

$$3x = 2x - 2$$

$$x = -2$$

$$y^2 = 1 \Rightarrow y = \pm 1$$

P(-2,1)

Q(-2,-1)

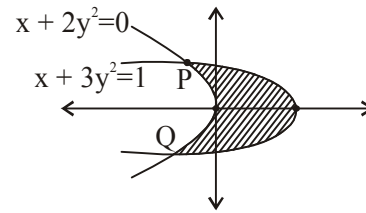
$$I = \int_{y_1}^{y_2} (x_2 - x_1) dy$$

$$= 2 \int_0^1 ((1 - 3y^2) - (-2y^2)) dy$$

$$= 2 \int_0^1 (1 - y^2) dy$$

$$= 2 \left(y - \frac{y^3}{3} \right)_0^1$$

$$= \frac{4}{3}$$



34. The area of the region bounded by the parabola $(y - 2)^2 = x - 1$, the tangent to the parabola at the point $(2, 3)$ and the x-axis is : [AIEEE2009]

परवलय $(y - 2)^2 = x - 1$, परवलय के बिन्दु $(2, 3)$ पर खींची गई स्पर्श रेखा तथा x-अक्ष द्वारा परिबद्ध क्षेत्र का क्षेत्रफल है : [AIEEE2009]

- (A) 3 (B) 6 (C) 9 (D) 12

Ans. C

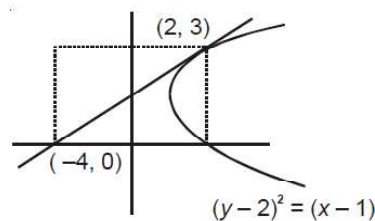
Sol. The equation of tangent at $(2, 3)$ to the given parabola is $x = 2y - 4$

$$\text{Required area} = \int_0^3 \left\{ (y - 2)^2 + 1 - 2y + 4 \right\} dy$$

$$= \left[\frac{(y - 2)^3}{3} - y^2 + 5y \right]_0^3$$

$$= \frac{1}{3} - 9 + 15 + \frac{8}{3}$$

$$= 9 \text{ sq. units}$$



35. The area (in square units) bounded by the curves $y = \sqrt{x}$, $2y - x + 3 = 0$ x-axis and lying in the first quadrant is: [JEE(Main)2013]

वक्रों $y = \sqrt{x}$, $2y - x + 3 = 0$ तथा x-अक्ष से घिरे उस क्षेत्र, जो प्रथम चतुर्थांश में स्थित है, का (वर्ग इकाई में) क्षेत्रफल है :

[JEE(Main)2013]

(A) 9

(B) 36

(C) 18

(D) 27/4

Ans. A

Sol. $y = \sqrt{x}$... (1)

and $2y - x + 3 = 0$... (2)

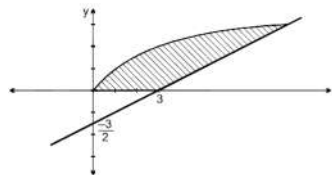
On solving both $y = -1, 3$

$$\text{Required area} = \int_0^3 \{(2y+3) - y^2\} dy$$

$$= y^2 + 3y - \frac{y^3}{3} \Big|_0^3$$

$$= 9 + 9 - 9$$

$$= 9$$



36. Let $g(x) = \cos x^2$, $f(x) = \sqrt{x}$, and α, β ($\alpha < \beta$) be the roots of the quadratic equation $18x^2 - 9\pi x + \pi^2 = 0$. Then the area (in sq. unit) bounded by the curve $y = (g \circ f)(x)$ and the lines $x = \alpha$, $x = \beta$ and $y = 0$ is :

[JEE(Main)2018]

माना $g(x) = \cos x^2$, $f(x) = \sqrt{x}$, तथा α, β ($\alpha < \beta$) द्विघाती समीकरण $18x^2 - 9\pi x + \pi^2 = 0$ के मूल हैं। तो वक्र $y = (g \circ f)(x)$ तथा $y = 0$ के बीच $x = \alpha$, $x = \beta$ तथा $y = 0$ द्वारा घिरे क्षेत्र का क्षेत्रफल (वर्ग इकाइयों में) है :

[JEE(Main)2018]

- (A) $\frac{1}{2}(\sqrt{3} - \sqrt{2})$ (B) $\frac{1}{2}(\sqrt{2} - 1)$ (C) $\frac{1}{2}(\sqrt{3} - 1)$ (D) $\frac{1}{2}(\sqrt{3} + 1)$

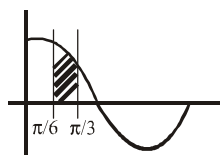
Ans. C

Sol. $g \circ f(x) = \cos x$

$$18x^2 - \pi x + \pi^2 = 0$$

$$(6x - \pi)(3x - \pi) = 0$$

$$x = \frac{\pi}{6}, x = \frac{\pi}{3}$$



$$A = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cos x \, dx = \frac{\sqrt{3} - 1}{2}$$

37. Let the straight line $x = b$ divide the area enclosed by $y = (1 - x^2)$, $y = 0$ and $x = 0$ into two parts R_1 ($0 \leq x \leq b$) and

R_2 ($b \leq x \leq 1$) such that $R_1 - R_2 = \frac{1}{4}$. Then b equals :

[JEE-2011]

यदि रेखा $x = b$ वक्रों $y = (1 - x^2)$, $y = 0$ तथा $x = 0$ को दो भागों में $R_1(0 \leq x \leq b)$ तथा $R_2(b \leq x \leq 1)$ में इस प्रकार विभाजित

द्वारा $R_1 - R_2 = \frac{1}{4}$ तो b का मान होगा :

[JEE-2011]

(A) $\frac{3}{4}$

(B) $\frac{1}{2}$

(C) $\frac{1}{3}$

(D) $\frac{1}{4}$

Ans. B

Sol. $\therefore R_1 - R_2 = \frac{1}{4}$

$$\Rightarrow \int_0^b (1-x)^2 dx - \int_b^1 (1-x)^2 dx = \frac{1}{4}$$

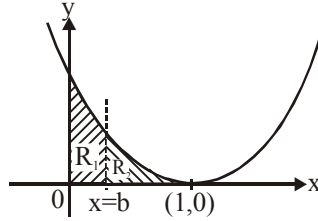
$$\Rightarrow -\left(\frac{(1-x)^3}{3}\right)_0^b + \left(\frac{(1-x)^3}{3}\right)_b^1 = \frac{1}{4}$$

$$\Rightarrow -\left\{\frac{(1-b)^3}{3} - \frac{1}{3}\right\} - \frac{(1-b)^3}{3} = \frac{1}{4}$$

$$\Rightarrow \frac{1}{3} - \frac{2}{3}(1-b)^3 = \frac{1}{4} \Rightarrow \frac{2}{3}(1-b)^3 = \frac{1}{12}$$

$$\Rightarrow (1-b)^3 = \frac{1}{8} \Rightarrow 1-b = \frac{1}{2}$$

$$\Rightarrow b = \frac{1}{2}$$



SHIFTING OF ORIGIN BASED QUESTIONS

38. The line $y = mx$ bisects the area enclosed by the curve $y = 1 + 4x - x^2$ and the lines $x = 0$, $x = 2$ and $y = 0$. Then the value of m is :

(A) $\frac{13}{6}$

(B) $\frac{6}{13}$

(C) $\frac{11}{6}$

(D) none

यदि रेखा $y = mx$, वक्र $y = 1 + 4x - x^2$ तथा रेखाएँ $x = 0$, $x = 2$ तथा $y = 0$ द्वारा परिबद्ध क्षेत्रफल को समद्विभाजित करती है, तो m का मान होगा :

(A) $\frac{13}{6}$

(B) $\frac{6}{13}$

(C) $\frac{11}{6}$

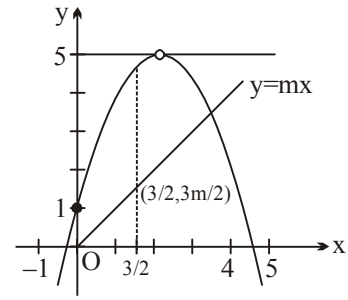
(D) कोई नहीं

Ans. A

Sol. $2 \cdot \frac{1}{2} \left(\frac{3}{2} \cdot \frac{3m}{2} \right) = \int_0^{3/2} (1 + 4x - x^2) dx$

$$\frac{9m}{4} = \left[x + 2x^2 - \frac{x^3}{3} \right]_0^{3/2} = \frac{3}{2} + \frac{9}{2} - \frac{27}{8} \cdot \frac{1}{3} = 6 - \frac{9}{8} = \frac{39}{8}$$

$$m = \frac{13}{6}$$



39. Area enclosed by the curve $|x - 2| + |y + 1| = 1$ is equal to :

- (A) 4 sq. units (B) 6 sq. units (C) 2 sq. units (D) 8 sq. units

वक्र $|x - 2| + |y + 1| = 1$ से घिरा हुआ क्षेत्रफल होगा :

- (A) 4 वर्ग इकाई (B) 6 वर्ग इकाई (C) 2 वर्ग इकाई (D) 8 वर्ग इकाई

Ans. C

Sol. $|x - 2| + |y - 1| = 1$

$$(0, 0) \rightarrow (2, 1)$$

$$|x| + |y| = 1$$

$$\text{Area} = \sqrt{2} \times \sqrt{2}$$

$$= 2$$

AREA REPRESENTED BY INEQUALITIES

40. The area of the region of the plane bounded by $\max(|x|, |y|) \leq 1$ and $xy \leq \frac{1}{2}$ is :

$\max(|x|, |y|) \leq 1$ तथा $xy \leq \frac{1}{2}$ द्वारा निरूपित क्षेत्र का क्षेत्रफल होगा :

- (A) $\frac{1}{2} + \ln 2$ (B) $\frac{31}{4}$ (C) $1 + 2\ln 2$ (D) $3 + \ln 2$

Ans. D

Sol. $\max(|x|, |y|) \leq 1$ $xy \leq \frac{1}{2}$

$$\text{Area} = 2 \times 2 - 2 \int_{\frac{1}{2}}^1 \left(1 - \frac{1}{2}x\right) dx$$

$$= 4 - 2 \left(x - \frac{1}{2} \ln x \right)_{\frac{1}{2}}^1$$

$$= 4 - 2 + 2 \left(\frac{1}{2} - \frac{1}{2} \ln \frac{1}{2} \right)$$

$$= 2 + 1 - \ln \frac{1}{2}$$

$$= 3 + \ln 2$$

41. Area of the region which consists of all the points satisfying the conditions $|x - y| + |x + y| \leq 8$ and $xy \geq 2$, is equal to:

(A) $4(7 - \ln 8)$ sq. units

(B) $4(9 - \ln 8)$ sq. units

(C) $2(7 - \ln 8)$ sq. units

(D) $2(9 - \ln 8)$ sq. units

प्रतिबंधों $|x - y| + |x + y| \leq 8$ तथा $xy \geq 2$ को संतुष्ट करने वाले सभी बिन्दु से बना हुआ कुल क्षेत्रफल होगा :

(A) $4; 7 - \ln 8$ वर्ग इकाई

(B) $4; 9 - \ln 8$ वर्ग इकाई

(C) $2; 7 - \ln 8$ वर्ग इकाई

(D) $2; 9 - \ln 8$ वर्ग इकाई

Ans. A

Sol. $|x - y| + |x + y| \leq 8$

$xy \geq 2$

$$\text{Required Area} = 2 \times \int_{\frac{1}{2}}^4 \left(4 - \frac{2}{x}\right) dx$$

$$= 2 \left(4x - 2 \ln x \right)_{\frac{1}{2}}^4$$

$$= 4 \left(8 - \ln 4 - 1 + \ln \left(\frac{1}{2} \right) \right)$$

$$= 4(7 - \ln 8)$$

42. A point P moves in xy plane in such a way that $[|x|] + [|y|] = 1$, where $[.]$ denotes the greatest integer function. Area of the region representing all possible positions of the point P is equal to :

- (A) 4 sq. units (B) 16 sq. units (C) $2\sqrt{2}$ sq. units (D) 8 sq. units

एक बिन्दु P xy तल में इस प्रकार गति करता है कि $[|x|] + [|y|] = 1$, जहाँ $[.]$ महत्तम पूर्णांक फलन है, तो P के सभी सम्भव मानों से घिरा क्षेत्रफल है:

- (A) 4 वर्ग इकाई (B) 16 वर्ग इकाई (C) $2\sqrt{2}$ वर्ग इकाई (D) 8 वर्ग इकाई

Ans. D

Sol. $[|x|] + [|y|] = 1$

1st Quadrant

$$[x] + [y] = 1$$

2nd Quadrant

$$[-x] + [y] = 1$$

3rd Quadrant

$$[-x] + [-y] = 1$$

4th Quadrant

$$[x] + [-y] = 1$$

$$\text{Area} = 2 \times 2 = 4$$

43. The area defined by $|y| \leq e^{-|x|} - \frac{1}{2}$ in cartesian co-ordinate system, is :

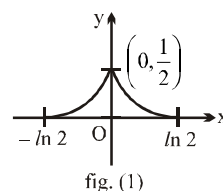
कार्तीय-निर्देशांक निकाय में $|y| \leq e^{-|x|} - \frac{1}{2}$ से परिभाषित क्षेत्रफल होगा :

- (A) $(4 - 2\ln 2)$ (B) $(4 - \ln 2)$ (C) $(2 - \ln 2)$ (D) $(2 - 2 \ln 2)$

Ans. D

Sol. For $y \geq 0$,

$$\therefore y = e^{-|x|} - \frac{1}{2} = \begin{cases} e^{-x} - \frac{1}{2} & \text{if } x \geq 0 \\ e^x - \frac{1}{2} & \text{if } x < 0 \end{cases} \quad \dots(1)$$

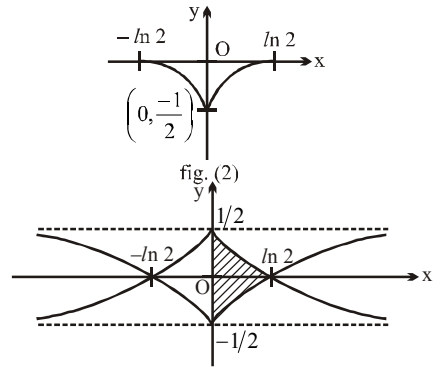


and for $y < 0$,

$$-y = \frac{1}{2} - e^{-|x|} = \begin{cases} \frac{1}{2} - e^{-x} & \text{if } x \geq 0 \\ \frac{1}{2} - e^x & \text{if } x < 0 \end{cases} \quad \dots(2)$$

Combining (1) and (2) we have the required area given as

$$A = 4 \int_0^{\ln 2} \left(e^{-x} - \frac{1}{2} \right) dx = 4 \left(\frac{1}{2} - \frac{1}{2} \ln 2 \right) = (2 - 2 \ln 2)$$

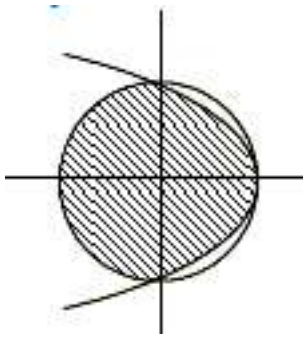


44. The area of the region described by $A = \{(x, y) : x^2 + y^2 \leq 1 \text{ and } y^2 \leq 1 - x\}$ is : **[JEE(Main)2014]**

$A = \{(x, y) : x^2 + y^2 \leq 1 \text{ तथा } y^2 \leq 1 - x\}$ के द्वारा प्रदत्त क्षेत्र का क्षेत्रफल है : **[JEE(Main)2014]**

- (A) $\frac{\pi}{2} - \frac{2}{3}$ (B) $\frac{\pi}{2} + \frac{2}{3}$ (C) $\frac{\pi}{2} + \frac{4}{3}$ (D) $\frac{\pi}{2} - \frac{4}{3}$

Ans. C



Sol.

$$\text{Required area} = \pi - 2 \int_0^1 (\sqrt{1-x^2} - \sqrt{1-x}) dx$$

$$= \pi - 2 \left[\frac{x\sqrt{1-x^2}}{2} + \frac{1}{2} \sin^{-1} x + \frac{2}{3} (1-x)^{3/2} \right]_0^1 = \pi - 2 \left[\frac{\pi}{4} - \frac{2}{3} \right] = \frac{\pi}{2} + \frac{4}{3}$$

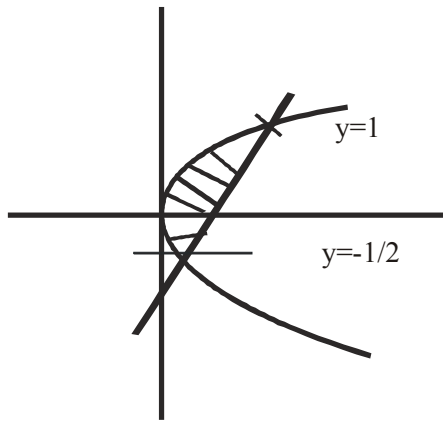
45. The area (in sq. units) of the region described by $[(x, y) : y^2 \leq 2x \text{ and } y \geq 4x - 1]$ is : **[JEE(Main)2015]**

$[(x, y) : y^2 \leq 2x \text{ तथा } y \geq 4x - 1]$ द्वारा परिभाषित क्षेत्र का क्षेत्रफल (वर्ग इकाईयों) में है : **[JEE(Main)2015]**

- (A) $\frac{1}{32}$ (B) $\frac{5}{64}$ (C) $\frac{15}{64}$ (D) $\frac{9}{32}$

Ans. D

Sol. $y^2 \leq 2x$ $y \geq 4x - 1$



$$\text{Req Area} = \int_{-\frac{1}{2}}^1 \left(\frac{y+1}{4} - \frac{y^2}{2} \right) dy$$

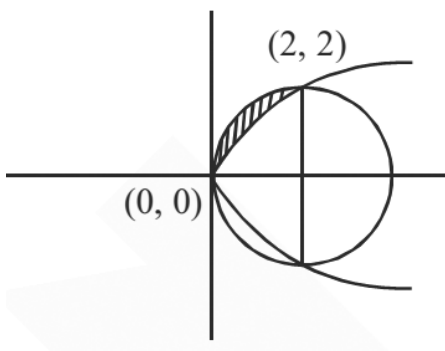
$$= \frac{9}{32}$$

46. The area (in sq. units) of the region $\{(x, y) : y^2 \geq 2x \text{ and } x^2 + y^2 \leq 4x, x \geq 0, y \geq 0\}$ is: **[JEE(Main)2016]**

क्षेत्र $\{(x, y) : y^2 \geq 2x \text{ तथा } x^2 + y^2 \leq 4x, x \geq 0, y \geq 0\}$ का क्षेत्रफल (वर्ग इकाइयों में) है: **[JEE(Main)2016]**

- (A) $\pi - \frac{4\sqrt{2}}{3}$ (B) $\frac{\pi}{2} - \frac{2\sqrt{2}}{3}$ (C) $\pi - \frac{4}{3}$ (D) $\pi - \frac{8}{3}$

Ans. D



Sol.

$$\text{Area} = \frac{1}{4} (\text{Area of circle}) - \int_0^2 \sqrt{2x} dx$$

$$= \pi - \frac{8}{3}$$

47. The area (in sq. units) of the region $\{(x, y) : x \geq 0, x + y \leq 3, x^2 \leq 4y \text{ and } y \leq 1 + \sqrt{x}\}$ is:

[JEE(Main)2017]

क्षेत्र $\{(x, y) : x \geq 0, x + y \leq 3, x^2 \leq 4y \text{ तथा } y \leq 1 + \sqrt{x}\}$ का क्षेत्रफल होगा :

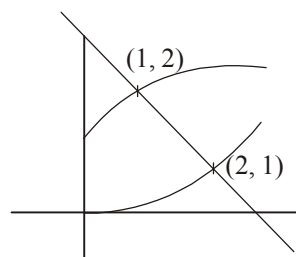
[JEE(Main)2017]

- (A) $\frac{7}{3}$ (B) $\frac{5}{2}$ (C) $\frac{59}{12}$ (D) $\frac{3}{2}$

Ans. B

Sol.
$$\text{Area} = \int_0^1 \left\{ (1 + \sqrt{x}) - \frac{x^2}{4} \right\} dx + \int_1^2 \left\{ (3 - x) - \frac{x^2}{4} \right\} dx$$

$$= \frac{5}{2}$$



DETERMINATION OF UNKNOWN PARAMETERS

48. If A is the area between the curve $y = \sin x$ and x-axis in the interval $[0, \pi/2]$, then the area between $y = \sin 2x$ and x-axis in this interval will be :

- (A) A (B) 2A (C) A/2 (D) None of these

यदि वक्र $y = \sin x$ तथा x-अक्ष के मध्य अन्तराल $[0, \pi/2]$ में क्षेत्रफल A हो, तो $y = \sin 2x$ तथा x-अक्ष के मध्य इसी अन्तराल में क्षेत्रफल

- (A) A (B) 2A (C) A/2 (D) इनमें से कोई नहीं

Ans. A

Sol. $y = \sin x$

$$A = \int_0^{\pi/2} \sin x dx$$

$$= -\cos x \Big|_0^{\pi/2}$$

$$= 1$$

$$A' = \int_0^{\pi/2} \sin 2x dx$$

$$= -\frac{\cos 2x}{2} \Big|_0^{\pi/2}$$

$$= -\frac{1}{2}[-1-1] = 1 = A$$

49. If $(a, 0)$; $a > 0$ is the point where the curve $y = \sin 2x - \sqrt{3} \sin x$ cuts the x-axis first, A is the area bounded by this part of the curve, the origin and the positive x-axis, then :

यदि $(a, 0)$; $a > 0$ वह बिन्दु है जहाँ वक्र $y = \sin 2x - \sqrt{3} \sin x$, x-अक्ष को प्रथम बार काटता है तथा वक्र के इस भाग, मूलबिन्दु तथा x-अक्ष द्वारा घिरा क्षेत्रफल A है, तब :

(A) $4A + 8 \cos a = 7$ (B) $4A + 8 \sin a = 7$ (C) $4A - 8 \sin a = 7$ (D) $4A - 8 \cos a = 7$

Ans. A

Sol. $(a, 0)$ lies on the given curve

$$\therefore 0 = \sin 2a - \sqrt{3} \sin a \Rightarrow \sin a = 0 \text{ or } \cos a = \sqrt{3}/2$$

$$\Rightarrow a = \frac{\pi}{6} \text{ (as } a > 0 \text{ and the first point of intersection with positive X-axis)}$$

$$\begin{aligned} \text{and } A &= \int_0^{\pi/6} (\sin 2x - \sqrt{3} \sin x) dx = \left(-\frac{\cos 2x}{2} + \sqrt{3} \cos x \right)_0^{\pi/6} \\ &= \left(-\frac{1}{4} + \frac{3}{2} \right) - \left(-\frac{1}{2} + \sqrt{3} \right) = \frac{7}{4} - \sqrt{3} = \frac{7}{4} - 2 \cos a \Rightarrow 4A + 8 \cos a = 7 \end{aligned}$$

AREA BOUNDED BY INVERSE OF A FUNCTION

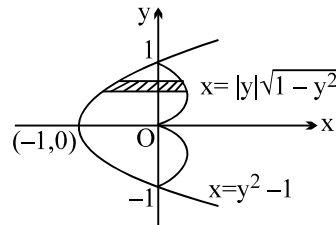
50. Area of the region enclosed between the curves $x = y^2 - 1$ and $x = |y| \sqrt{1 - y^2}$ is :

वक्रों $x = y^2 - 1$ तथा $x = |y| \sqrt{1 - y^2}$ द्वारा परिबद्ध क्षेत्रफल होगा :

(A) 1 (B) 4/3 (C) 2/3 (D) 2

Ans. D

$$\begin{aligned} \text{Sol. } A &= 2 \int_0^1 [y\sqrt{1-y^2} - (y^2-1)] dy \\ &= 2 \end{aligned}$$



DETERMINATION OF FUNCTION

51. The area bounded by the curve $y = f(x)$, the x -axis & the ordinates $x = 1$ & $x = b$ is $(b-1) \sin(3b+4)$. Then $f(x)$ is :

(A) $(x-1) \cos(3x+4)$ (B) $\sin(3x+4)$
(C) $\sin(3x+4) + 3(x-1) \cdot \cos(3x+4)$ (D) none

यदि वक्र $y = f(x)$, x -अक्ष तथा कोटि $x = 1$ तथा $x = b$ द्वारा परिबद्ध क्षेत्रफल $(b-1) \sin(3b+4)$ है, तो $f(x)$ होगा :

(A) $(x-1) \cos(3x+4)$ (B) $\sin(3x+4)$
(C) $\sin(3x+4) + 3(x-1) \cdot \cos(3x+4)$ (D) कोई नहीं

Ans. C

Sol.
$$\int_1^b f(x) dx = (b-1) \sin(3b+4)$$

Apply Leibnitz

$$f(b) = (b-1) \cos(3b+4) + \sin(3b+4)$$

52. If the area bounded by the x -axis, curve $y = f(x)$ and the lines $x = 1$, $x = b$ is equal to $\sqrt{b^2+1} - \sqrt{2}$ for all $b > 1$, then $f(x)$ is :

यदि x -अक्ष, वक्र $y = f(x)$ तथा रेखाओं $x = 1$, $x = b$ के द्वारा परिबद्ध क्षेत्रफल $\sqrt{b^2+1} - \sqrt{2}$; सभी $b > 1$ के लिए हो, तो $f(x)$ होगा :

(A) $\sqrt{x-1}$ (B) $\sqrt{x+1}$ (C) $\sqrt{x^2+1}$ (D) $\frac{x}{\sqrt{1+x^2}}$

Ans. D

Sol.
$$\int_1^b f(x) dx = \sqrt{b^2+1} - \sqrt{2}$$

$$f(b) = \frac{2b}{2\sqrt{b^2+1}} = \frac{b}{\sqrt{b^2+1}}$$

53. Let $f(x)$ be a non-negative continuous function such that the area bounded by the curve $y = f(x)$, x -axis and the ordinates $x = \frac{\pi}{4}$ and $x = \beta > \frac{\pi}{4}$ is $\left(\beta \sin \beta + \frac{\pi}{4} \cos \beta + \sqrt{2} \beta \right)$. Then $f\left(\frac{\pi}{2}\right)$ is :

माना $f(x)$ एक इस प्रकार का अकृणात्मक सतत् फलन है, कि वक्र $y = f(x)$, x -अक्ष तथा रेखाओं $x = \frac{\pi}{4}$ तथा

$x = \beta > \frac{\pi}{4}$ से आबद्ध क्षेत्र का क्षेत्रफल $\left(\beta \sin \beta + \frac{\pi}{4} \cos \beta + \sqrt{2} \beta \right)$ है, तो $f\left(\frac{\pi}{2}\right)$ का मान है :

- (A) $\left(\frac{\pi}{4} + \sqrt{2} - 1\right)$ (B) $\left(\frac{\pi}{4} - \sqrt{2} + 1\right)$ (C) $\left(1 - \frac{\pi}{4} - \sqrt{2}\right)$ (D) $\left(1 - \frac{\pi}{4} + \sqrt{2}\right)$

Ans. D

Sol.
$$\int_{\frac{\pi}{4}}^{\beta} f(x) dx = \beta \sin \beta + \frac{\pi}{4} \cos \beta + \sqrt{2} \beta$$

$$f(\beta) = \beta \cos \beta + \sin \beta - \frac{\pi}{4} \sin \beta + \sqrt{2}$$

$$f\left(\frac{\pi}{2}\right) = 0 + 1 - \frac{\pi}{4} + \sqrt{2}$$

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

AREA BOUNDED BETWEEN TWO CURVES

1. Find the area bounded on the right by the line $x + y = 2$, on the left by the parabola $y = x^2$ and below by the x -axis in first quadrant.

क्षेत्रफल ज्ञात कीजिए, जो दांयी ओर से सरल रेखा $x + y = 2$ से, बांयी ओर से परवलय $y = x^2$ से, नीचे की तरफ से x -अक्ष से परिबद्ध हो (प्रथम चतुर्थांश में)।

Ans. $5/6$ sq. units

Sol. Line $x + y = 2$ and $y = x^2$

POI $x + y = 2$

$$y = x^2$$

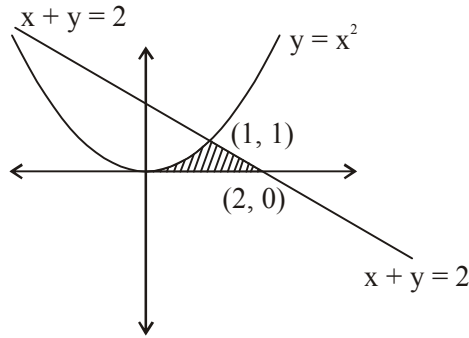
$(0, 0)$ and $(1, 1)$

$$\text{Area} = \int_0^1 y_1 dx + \int_1^0 y_2 dx$$

$$A = \int_0^1 x^2 dx + \int_1^2 (2 - x) dx$$

$$= \left. \frac{x^3}{3} \right|_0^1 + 2x - \frac{x^2}{2} \Big|_1^2$$

$$= \frac{1}{3} + 4 - 2 - 2 + \frac{1}{2} = \frac{5}{6}$$

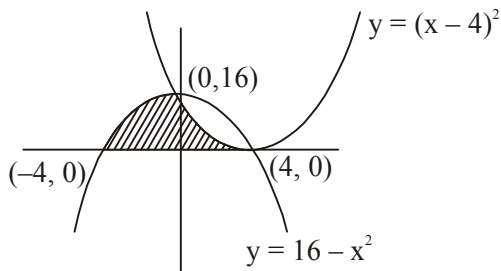


2. Find the area of the region bounded by curves $f(x) = (x - 4)^2$, $g(x) = 16 - x^2$ and the x -axis.

वक्रों $f(x) = (x - 4)^2$, $g(x) = 16 - x^2$ और x -अक्ष से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

Ans. 64

Sol. $f(x) = (x - 4)^2$ $g(x) = 16 - x^2$ x -axis



$$A = \int_{-4}^0 (16 - x^2) dx + \int_0^4 (x - 4)^2 dx$$

$$= 16x - \frac{x^3}{3} \Big|_{-4}^0 + \frac{(x-4)^3}{3} \Big|_0^4$$

$$= 64 + \frac{16}{3} - \frac{16}{3} = 64$$

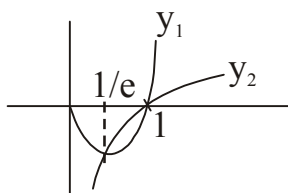
3. Compute the area of the region bounded by the curves $y = e \cdot x \cdot \ln x$ & $y = \ln x / (e \cdot x)$ where $\ln e = 1$.

यहाँ $y = e \cdot x \cdot \ln x$ और $y = \ln x / (e \cdot x)$ जहाँ $\ln e = 1$ से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

Ans. $(e^2 - 5)/4 e$ sq. units

Sol. $y_1 = e \cdot x \cdot \ln x$

$$y_2 = \frac{\ln x}{ex}$$



$$A = \int_{\frac{1}{e}}^1 \left(\frac{\ln x}{ex} - ex \ln x \right) dx$$

$$= \frac{1}{e} \left(\frac{(\ln x)^2}{2} \right) \Big|_{\frac{1}{e}}^1 - e \left[\frac{x^2}{2} \ln x - \frac{x^2}{4} \right] \Big|_{\frac{1}{e}}^1$$

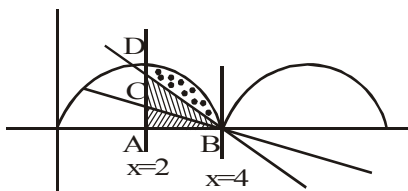
$$\begin{aligned}
&= \frac{1}{e} \left(-\frac{1}{2} \right) - e \left[-\frac{1}{4} + \frac{1}{2e^2} + \frac{1}{4e^2} \right] \\
&= -\frac{1}{2e} + \frac{e}{4} - \frac{3}{4e} \\
&= \frac{e}{4} - \frac{5}{4e} \\
&= \frac{e^2 - 5}{4e}
\end{aligned}$$

4. A figure is bounded by the curves $y = \left| \sqrt{2} \sin \frac{\pi x}{4} \right|$, $y = 0$, $x = 2$ & $x = 4$. At what angles to the positive x -axis straight lines must be drawn through $(4, 0)$ so that these lines partition the figure into three parts of the same size.

एक क्षेत्र वक्रों $y = \left| \sqrt{2} \sin \frac{\pi x}{4} \right|$, $y = 0$, $x = 2$ और $x = 4$ से परिबद्ध है। तब x -अक्ष की धनात्मक दिशा किस कोण से सरल रेखा जो कि $(4, 0)$ से जाती है खींची जाए की वह क्षेत्र को समान आकार के तीन भागों में विभाजित कर दे।

Ans. $\pi - \tan^{-1} \frac{2\sqrt{2}}{3\pi}$; $\pi - \tan^{-1} \frac{4\sqrt{2}}{3\pi}$

Sol. Let equation of line is $y = mx - 4m$



$$A = \int_2^4 \sqrt{2} \sin \frac{\pi x}{4} dx = \left[-\sqrt{2} \frac{4}{\pi} \cos \frac{\pi x}{4} \right]_2^4 = \frac{4\sqrt{2}}{\pi} \dots (i)$$

$$\text{Also area of } \triangle ABC = \frac{1}{2} \cdot 2 \cdot (-2m_1) = -2m_1 \dots (ii)$$

from (i) and (ii)

$$-2m_1 = \frac{4\sqrt{2}}{3\pi} \Rightarrow m_1 = \frac{-2\sqrt{2}}{3\pi}$$

$$\Rightarrow \tan(\pi - \theta_1) = \frac{-2\sqrt{2}}{3\pi} \Rightarrow \pi - \theta_1 = \tan^{-1} \frac{2\sqrt{2}}{3\pi}$$

$$\Rightarrow q_1 = \pi - \tan^{-1} \frac{2\sqrt{2}}{3\pi} \text{ or } \frac{1}{2} \cdot (2) \cdot (-2m_2) = \frac{8\sqrt{2}}{3\pi}$$

$$\Rightarrow m_2 = \frac{-4\sqrt{2}}{3\pi} \Rightarrow \tan(\pi - \theta_2) = \frac{-4\sqrt{2}}{3\pi}$$

$$\Rightarrow q_2 = \pi - \tan^{-1} \frac{4\sqrt{2}}{3\pi}$$

5. Find the area bounded by the curves $y = \sqrt{1-x^2}$ and $y = x^3 - x$. Also find the ratio in which the y-axis divided this area.

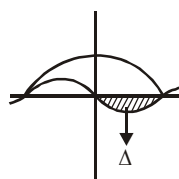
वक्रों $y = \sqrt{1-x^2}$ और $y = x^3 - x$ से परिबद्ध क्षेत्रफल ज्ञात कीजिए। साथ ही वह अनुपात भी बताईये जबकि y-अक्ष क्षेत्रफल को विभाजित करता है।

Ans. $\frac{\pi}{2}; \frac{\pi-1}{\pi+1}$

Sol. $\Delta = \left| \int_0^1 (x^3 - x) dx \right| = \frac{1}{4}$

$$\text{Area} = \frac{\pi}{4} + \frac{1}{4} + \frac{\pi}{4} - \frac{1}{4} = \frac{\pi}{2}$$

$$\text{required ratio} = \frac{\frac{\pi}{4} + \frac{1}{4}}{\frac{\pi}{4} - \frac{1}{4}}$$



6. The line $3x + 2y = 13$ divides the area enclosed by the curve, $9x^2 + 4y^2 - 18x - 16y - 11 = 0$ into two parts. Find the ratio of the larger area to the smaller area.

यदि सरल रेखा $3x + 2y = 13$, वक्र $9x^2 + 4y^2 - 18x - 16y - 11 = 0$ के बन्द क्षेत्र को दो भागों में विभाजित करती है तब बड़े आकार के क्षेत्रफल तथा छोटे क्षेत्र के क्षेत्रफलों का अनुपात ज्ञात कीजिए।

Ans. $\frac{3\pi + 2}{\pi - 2}$

Sol. $3x + 2y = 13$

$$9x^2 + 4y^2 - 18x - 16y - 11 = 0$$

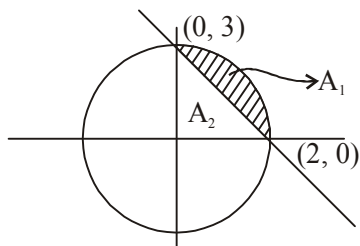
$$9(x-1)^2 + 4(y-2)^2 - 36 = 0$$

$$\frac{(x-1)^2}{4} + \frac{(y-2)^2}{9} = 1$$

$$x-1=x \quad y-2=y$$

$$\frac{x^2}{4} + \frac{y^2}{9} = 1 \text{ and } 3(x+1) + 2(y+2) = 13$$

$$3x + 2y = 6$$



$$A_2 = \frac{1}{2} \times 2 \times 3 = 3$$

$$\begin{aligned} \text{Area of ellipse} &= \pi ab = \pi \times 2 \times 3 \\ &= 6\pi \end{aligned}$$

$$A_1 = \frac{6\pi}{4} - 3$$

$$= \frac{3\pi - 6}{2}$$

Non-shaded ellipse area

$$= \frac{3}{4} \times 6\pi + 3 = \frac{9\pi + 6}{2}$$

$$\text{Ratio} = \frac{\frac{9\pi + 6}{2}}{\frac{3\pi - 6}{2}} = \frac{3\pi + 2}{\pi - 2} \text{ Ans.}$$

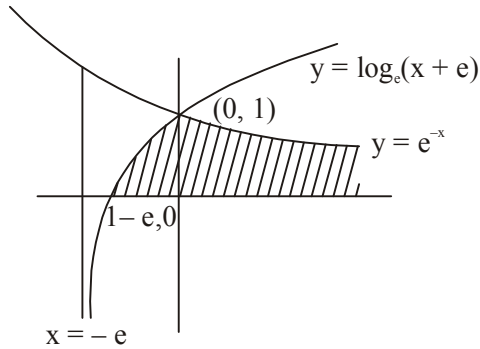
7. Find the area enclosed between the curves : $y = \log_e(x+e)$, $x = \log_e(1/y)$ & the x-axis.

चक्रों $y = \log_e(x+e)$, $x = \log_e(1/y)$ और x-अक्ष से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

Ans. 2 sq. units

Sol. $y = \log_e(x + e)$ and x-axis

$$x = \log_e(y_y) = y = e^{-x}$$



$$\int_{1-e}^0 \log_e(x + e) dx + \int_0^{\infty} e^{-x} dx$$

$$= (x + e) \log_e \left(\frac{x + e}{e} \right) \Big|_{1-e}^0 + -e^{-x} \Big|_0^{\infty}$$

$$A = 0 - 1(-1) + 1 = 2 \text{ Ans.}$$

8. Compute the area of the curvilinear triangle bounded by the y-axis & the curve, $y = \tan x$ & $y = (2/3)\cos x$.

वक्रों $y = \tan x$ और $y = (2/3)\cos x$ और y-अक्ष से परिबद्ध वक्रिय त्रिभुज का क्षेत्रफल ज्ञात कीजिए।

Ans. $\frac{1}{3} + \ell n \left(\frac{\sqrt{3}}{2} \right)$ sq. units

Sol. $y = \tan x$ $y = \frac{2}{3} \cos x$ y-axis

POI

$$\tan x = \frac{2}{3} \cos x$$

$$3 \tan x = 2 \cos x$$

$$3 \sin x = 2 \cos^2 x$$

$$3 \sin x = 2 - 2 \sin^2 x$$

$$2 \sin^2 x + 3 \sin x - 2 = 0$$

$$2 \sin^2 x + 4 \sin x - \sin x - 2 = 0$$

$$(2 \sin x - 1)(\sin x + 2) = 0$$

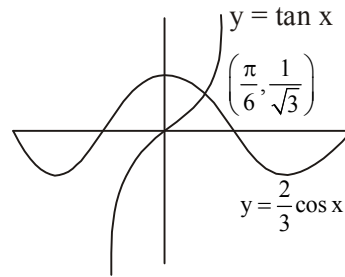
$$\sin x = -1$$

$$x = \frac{\pi}{6}$$

$$A = \int_0^{\frac{\pi}{6}} \left(\frac{2}{3} \cos x - \tan x \right) dx$$

$$= \frac{2}{3} \sin x - |\ln|\sec x||_0^{\frac{\pi}{6}}$$

$$= \frac{2}{3} \times \frac{1}{2} - |\ln|\frac{2}{\sqrt{3}}|| = \frac{1}{3} + |\ln|\frac{\sqrt{3}}{2}||$$



9. Find the area bounded by the curve $y = x e^{-x}$; $xy = 0$ and $x = c$ where c is the x -coordinate of the curve's inflection point.

वक्रों $y = x e^{-x}$; $xy = 0$ और $x = c$ से परिबद्ध क्षेत्रफल ज्ञात कीजिए। जहाँ c वक्र के नति परिवर्तन बिन्दु का x -निर्देशांक है।

Ans. $1 - 3e^{-2}$

Sol. $y = x e^{-x}$

$$y' = e^{-x} - x e^{-x} = (1 - x)e^{-x} \uparrow \text{ for } x < 1$$

$$y'' = -e^{-x} - [e^{-x} - x e^{-x}] = e^{-x}[-1 - 1 + x]$$

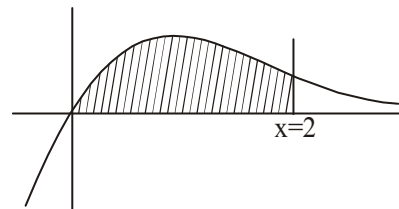
$$= (x - 2)e^{-x}$$

$$\text{for point of inflection } y'' = 0 \Rightarrow x = 2$$

$$A = \int_0^2 x e^{-x} dx = -x e^{-x} \Big|_0^2 + \int_0^2 e^{-x}$$

$$= (-2 e^{-2}) - (e^{-x})_0^2$$

$$= -2 e^{-2} - (e^{-2} - 1) = 1 - e^{-2} - 2e^{-2} = 1 - 3e^{-2}$$



10. Find the area bounded by the curve $y = x e^{-x^2}$, the x-axis, and the line $x = c$ where $y(c)$ is maximum.

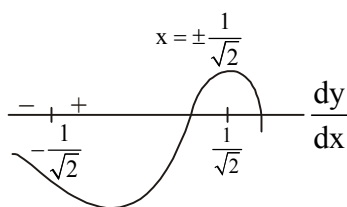
वक्र $y = x e^{-x^2}$, x-अक्ष तथा रेखा $x = c$ परिवर्द्ध क्षेत्रफल ज्ञात कीजिए। (जहाँ $y(c)$ अधिकतम है।)

Ans. $\frac{1}{2}(1 - e^{-1/2})$

Sol. $y = x e^{-x^2}$, x-axis $x = c$

$$\frac{dy}{dx} = -2x^2 e^{-x^2} + e^{-x^2} = 0$$

$$x = \pm \frac{1}{\sqrt{2}}$$



$$\left. \frac{dy}{dx} \right|_{\max} \text{ at } x = \frac{1}{\sqrt{2}} \quad c = \frac{1}{\sqrt{2}}$$

$$A = \int_0^{\frac{1}{\sqrt{2}}} x e^{-x^2} dx$$

$$x^2 = t \Rightarrow 2x dx = dt$$

$$= \int_0^{\frac{1}{2}} e^{-t} \frac{dt}{2}$$

$$= \frac{1}{2} \left[-e^{-t} \right]_0^{\frac{1}{2}}$$

$$= \frac{1}{2} \left(1 - e^{-\frac{1}{2}} \right) \text{ Ans.}$$

11. Consider a circle $x^2 + (y - 1)^2 = 1$ and the parabola $y = -\frac{x^2}{4}$. The common tangents to the two curves constitute a triangle ABC, the point A and B being on the x-axis and C on the y-axis. CA produced touches the parabola at P and CB produced touches the parabola at Q.
- (a) Find the equation of the common tangent BC.
- (b) Find the area of the portion between the upper arc of the circle and the common tangents QC and PC.
- (c) Find the area enclosed by the parabola $y = -\frac{x^2}{4}$, the x-axis and the lines AP and BQ.

माना वृत्त $x^2 + (y - 1)^2 = 1$ और परवलय $y = -\frac{x^2}{4}$ दिया गया है। यदि वक्रों की उभयनिष्ठ स्पर्शरेखाएँ एक त्रिभुज ABC का निर्माण करता है, जहाँ बिन्दु A और B, x-अक्ष पर हैं और C, y-अक्ष पर, यदि CA का बढ़ाया भाग परवलय को P पर स्पर्श करता है और CB का बढ़ाया भाग परवलय को Q पर स्पर्श करता है।

- (a) उभयनिष्ठ स्पर्शरेखा BC ज्ञात कीजिए।
- (b) वृत्त के ऊपरी चाप और उभयनिष्ठ स्पर्श रेखाओं QC और PC से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।
- (c) परवलय $y = -\frac{x^2}{4}$, x-अक्ष और रेखाओं AP और BQ से परिबद्ध क्षेत्रफल ज्ञात कीजिए।

Ans. (a) $\sqrt{3}x - y + 3 = 0$ (b) $\left(\sqrt{3} - \frac{\pi}{3}\right)$ (c) $\sqrt{3}$

Sol. $x^2 + (y - 1)^2 = 1$ and $x^2 = -4y$

Common tangent at parabola

$$x = \frac{y}{m} - m$$

$$y = mx + m^2$$

COT from (0, 1)

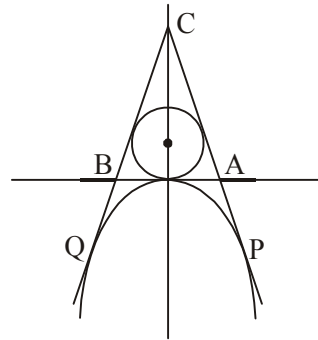
$$\left| \frac{1 - m^2}{\sqrt{1 + m^2}} \right| = 1$$

$$m^4 - 2m^2 + 1 = 1 + m^2$$

$$m^4 = m^2$$

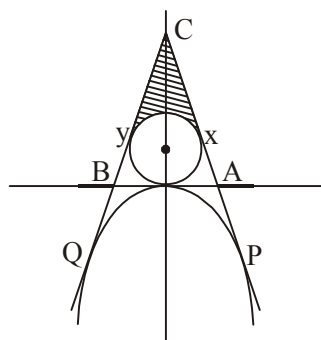
$$m = 0, m = \sqrt{3}, m = -\sqrt{3}$$

$$AB : y = 0 \quad BC : y = \sqrt{3}x + 3 \quad AC : y = -\sqrt{3}x + 3$$



$$A(-\sqrt{3}, 0) \quad B(\sqrt{3}, 0) \quad C(0, 3)$$

Point of contact on circle



12. If $f(x)$ is monotonic in (a, b) then prove that the area bounded by the ordinates at $x = a$; $x = b$; $y = f(x)$ and $y = f(c)$, $c \in (a, b)$ is minimum when $c = \frac{a+b}{2}$. Hence if the area bounded by the graph of $f(x) = \frac{x^3}{3} - x^2 + a$, the straight lines $x = 0$, $x = 2$ and the x -axis is minimum then find the value of 'a'.

यदि $f(x)$; (a, b) में एकदिष्ट फलन है, तो सिद्ध कीजिये कि $x = a$; $x = b$; $y = f(x)$ और $y = f(c)$, $c \in (a, b)$ से परिबद्ध क्षेत्रफल न्यूनतम होगा यदि $c = \frac{a+b}{2}$ । अब यदि वक्र $f(x) = \frac{x^3}{3} - x^2 + a$, सरल रेखा $x = 0$, $x = 2$ और x -अक्ष से परिबद्ध क्षेत्रफल न्यूनतम है, तो 'a' का मान ज्ञात कीजिये।

Ans. $a = \frac{2}{3}$

Sol. $f(x) \uparrow (a, b)$

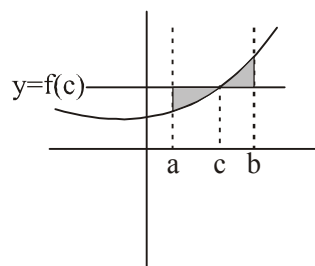
Shaded area

$$A = \int_a^c (f(c) - f(x)) dx + \int_c^b (f(x) - f(c)) dx$$

$$A = f(c)(c-a) - \int_c^a f(x) dx + \int_c^b f(x) dx - f(c)(b-c)$$

$$A = f(c)(2c-a-b) + \underbrace{\int_c^a f(x) dx + \int_c^b f(x) dx}_{+ive}$$

$$f(c)(2c-a-b) = 0$$



$$c = \frac{a+b}{2}$$

$$f(x) = \frac{x^3}{3} - x^2 + a$$

$$f'(x) = x^2 - 2x$$

$$-x = 0, x = 2$$

Area will be minimum when graph will be symmetric about root α .

$$\alpha = \frac{0+2}{2} = 1$$

$$f(1) = 0$$

$$-\frac{2}{3} + a = 0$$

$$a = \frac{2}{3}$$

13. Find the area of the region bounded by the curves $y = x^2$, $y = |2 - x^2|$ and $y = 2$, which lies to the right of the line $x = 1$.

वक्रों $y = x^2$, $y = |2 - x^2|$ और रेखा $y = 2$ से परिबद्ध उस क्षेत्र का क्षेत्रफल ज्ञात कीजिये जो कि रेखा $x = 1$ के दायीं ओर है।

Ans. $\left(\frac{20}{3} - 4\sqrt{2}\right)$ sq. units

Sol. The given curves are $y = x^2$

which is an upward parabola with vertex at $(0, 0)$

$$y = |2 - x^2|$$

$$\text{or } y = \begin{cases} 2 - x^2 & \text{if } -\sqrt{2} < x < \sqrt{2} \\ x^2 - 2 & \text{if } x < -\sqrt{2} \text{ or } x > \sqrt{2} \end{cases}$$

$$\text{or } x^2 = -(y - 2); \quad -\sqrt{2} < x < \sqrt{2} \quad \dots (2)$$

a downward parabola with vertex at $(0, 2)$

$$x^2 = y + 2; \quad x < -\sqrt{2}, x > \sqrt{2} \quad \dots (3)$$

On upward parabola with vertex at $(0, -2)$

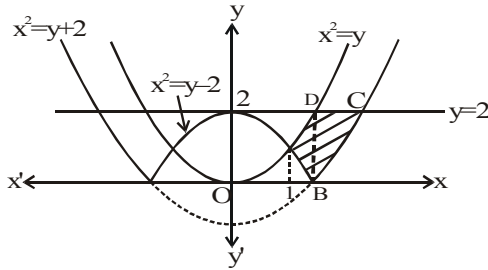
$$y = 2 \quad \dots (4)$$

Straight line parallel to x-axis

$$x = 1 \quad \dots (5)$$

Straight line parallel to y-axis

The graph of these curves is as follows.



$\therefore$ Required area = BCDEB

$$= \int_1^{\sqrt{2}} [x^2 - (2 - x^2)] dx + \int_{\sqrt{2}}^2 [2 - (x^2 - 2)] dx$$

$$= \int_1^{\sqrt{2}} (2x^2 - 2) dx + \int_{\sqrt{2}}^2 (4 - x^2) dx = \left(\frac{20}{3} - 4\sqrt{2} \right) \text{sq. units}$$

14. Find the area bounded by the curves $x^2 = y$, $x^2 = -y$ and $y^2 = 4x - 3$.

वक्रों $x^2 = y$, $x^2 = -y$ और $y^2 = 4x - 3$ से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिये।

Ans. $\frac{1}{3}$ sq. units

Sol. $x^2 = y$

$$x^2 = -y$$

$$y^2 = x \left(x - \frac{3}{4} \right)$$

POI

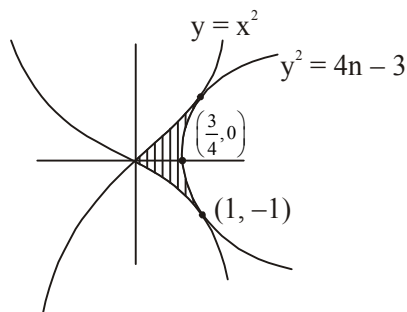
$$y^2 = 4x - 3 \text{ and } x^2 = y$$

$$x^4 = 4x - 3$$

$$x^4 - 4x + 3 = 0$$

$$(x - 1)(x^3 + x^2 + x - 3) = 0$$

$$(x - 1)(x - 1) \left(\underbrace{x^2 + 2x + 3}_{D < 0} \right) = 0$$



$$n = 1$$

$$A = 2 \left[\int_0^1 x^2 dx - \int_{\frac{3}{4}}^1 \sqrt{4x-3} dx \right]$$

$$= 2 \left[\frac{x^3}{3} \Big|_0^1 - \frac{2}{3} \frac{(4x-3)^{\frac{3}{2}}}{\frac{3}{2}} \Big|_{\frac{3}{4}}^1 \right]$$

$$= 2 \left(\frac{1}{3} - \frac{1}{6} \right)$$

$$= 2 \times \frac{1}{6} = \frac{1}{3}$$

15. Let $f(x)$ be a quadratic polynomial and a, b, c be distinct real numbers such that

$$\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix} \begin{bmatrix} f(-1) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 3a^2 + 3a \\ 3b^2 + 3b \\ 3c^2 + 3c \end{bmatrix}.$$

Let V be the point of maximum of the curve $y = f(x)$. If A and B are the points on this curve such that the curve meets the positive x -axis at A and the chord AB subtends a right angle at V , then find the area enclosed by the curve and the chord AB .

माना $f(x)$ एक वर्ग बहुपद है और a, b, c तीन अलग-अलग वास्तविक संख्याएँ हैं, इस प्रकार है कि

$$\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix} \begin{bmatrix} f(-1) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 3a^2 + 3a \\ 3b^2 + 3b \\ 3c^2 + 3c \end{bmatrix}.$$

माना $y = f(x)$ का उच्चिष्ठ बिन्दु V है। यदि A और B वक्र पर दो बिन्दु इस प्रकार हैं कि वक्र x -अक्ष को बिन्दु A पर मिलता है तथा जीवा AB , V पर समकोण बनाती है, तब वक्र और जीवा AB से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

Ans. $\frac{125}{3}$ sq. units

Sol. We have, $\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix} \begin{bmatrix} f(-1) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 3a^2 + 3a \\ 3b^2 + 3b \\ 3c^2 + 3c \end{bmatrix}$

$$\begin{aligned}\Rightarrow 4a^2f(-1) + 4af(1) + f(2) &= 3a^2 + 3a \\ 4b^2f(-1) + 4bf(1) + f(2) &= 3b^2 + 3b \\ 4c^2f(-1) + 4cf(1) + f(2) &= 3c^2 + 3c\end{aligned}$$

Consider the equation

$$\begin{aligned}4x^2f(-1) + 4xf(1) + f(2) &= 3x^2 + 3x \\ \text{or } [4f(-1) - 3]x^2 + [4f(1) - 3]x + f(2) &= 0\end{aligned}$$

Then clearly this equation is satisfied by

$$x = a, b, c$$

A quadratic equation satisfied by more than two values of x means it is an identity and hence

$$\begin{aligned}4f(-1) - 3 &= 0 \quad \Rightarrow \quad f(-1) = 3/4 \\ 4f(1) - 3 &= 0 \quad \Rightarrow \quad f(1) = 3/4 \\ f(2) &= 0 \quad \Rightarrow \quad f(2) = 0\end{aligned}$$

Let $f(x) = px^2 + qx + r$ [$f(x)$ being a quad. equation]

$$f(-1) = \frac{3}{4} \quad \Rightarrow \quad p - q + r = \frac{3}{4}$$

$$f(1) = \frac{3}{4} \quad \Rightarrow \quad p + q + r = \frac{3}{4}$$

$$f(2) = 0 \quad \Rightarrow \quad 4p + 2q + r = 0$$

Solving the above we get $q = 0, p = \frac{-1}{4}, r = 1$

$$\therefore f(x) = -\frac{1}{4}x^2 + 1$$

It's maximum value occur at $f'(x) = 0$

i.e., $x = 0$ then $f(x) = 1 \quad \therefore V(0, 1)$

$A(-2, 0)$ is the pt. where curve meet x -axis

Let B be the pt. $\left(h, \frac{4-h^2}{4}\right)$

$$\begin{aligned}\text{As } \angle AVB &= 90^\circ \\ m_{AV} \times m_{BV} &= -1\end{aligned}$$

$$\Rightarrow \frac{1}{2} \times \left(\frac{-h}{4}\right) = -1$$

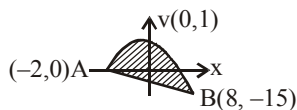
$$\Rightarrow h = 8 \quad \therefore B(8, -15)$$

Equation of chord AB is

$$y + 15 = \frac{0 - (-15)}{-2 - 8}(x - 8)$$

$$\Rightarrow 3x + 2y + 6 = 0$$

Required area is



the area of shaded region given by

$$= \int_{-2}^8 \left[\left(-\frac{x^2}{4} + 1 \right) - \left(\frac{-6-3x}{2} \right) \right] dx$$

$$= \frac{125}{3} \text{ sq. units.}$$

DETERMINATION OF UNKNOWN PARAMETERS

16. If the area enclosed by the parabolas $y = a - x^2$ and $y = x^2$ is $18\sqrt{2}$ sq. units. Find the value of 'a'.

यदि वक्रों $y = a - x^2$ और $y = x^2$ से परिबद्ध क्षेत्रफल $18\sqrt{2}$ वर्ग इकाई है। तब a का मान ज्ञात कीजिए।

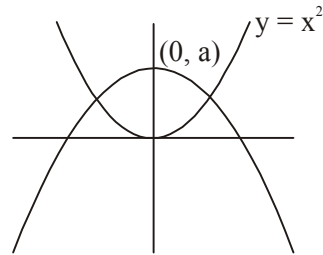
Ans. $a = 9$

Sol. $y = a - x^2$

$$y = x^2$$

$$a - x^2 = x^2$$

$$x^2 = \frac{a}{2}$$



$$A = 2 \int_0^{\sqrt{\frac{a}{2}}} (a - x^2 - x^2) dx$$

$$18\sqrt{2} = 2 \left[ax - \frac{2x^3}{3} \right]_0^{\sqrt{\frac{a}{2}}}$$

$$9\sqrt{2} = \frac{a^{\frac{3}{2}}}{\sqrt{2}} - \frac{2a^{\frac{3}{2}}}{3\sqrt{2}}$$

$$18 = \frac{3a^{\frac{3}{2}} - a^{\frac{3}{2}}}{3}$$

$$a^{\frac{3}{2}} = 27$$

$$a = 9$$

17. Find the values of m ($m > 0$) for which the area bounded by the line $y = mx + 2$ and $x = 2y - y^2$ is :
 (i) $9/2$ square units and (ii) minimum. Also find the minimum area.

m ($m > 0$) के मानों को ज्ञात कीजिए जबकि सरल रेखा $y = mx + 2$ और वक्र $x = 2y - y^2$ से परिबद्ध क्षेत्रफल :
 ; पद 9/2 वर्ग इकाई है (ii) न्यूनतम है। साथ ही न्यूनतम क्षेत्रफल ज्ञात कीजिए।

Ans. (i) $m = 1$ (ii) $m = \infty$; $A_{\min} = 4/3$

Sol. $y = mx + 2$; $m > 0$ and $x = 2y - y^2$

POI

$$y = mx + 2$$

$$x = \frac{y-2}{m} = 2y - y^2$$

$$y - 2 = m(y - 2)(-y)$$

$$y = -\frac{1}{m}, 2$$

$$x = -\frac{1-2m}{m^2}, 0$$

$$P\left(\frac{2m-1}{m^2}, -\frac{1}{3}\right) \text{ and } Q(0, 2)$$

$$\text{Area} = \int_{-\frac{1}{m}}^2 \left[(2y - y^2) - \left(\frac{y-2}{m} \right) \right] dy$$

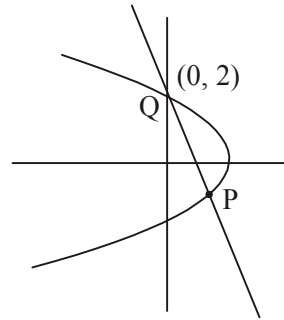
$$A = y^2 - \frac{y^2}{3} - \frac{y^2}{2m} + \frac{2y}{m} \Big|_{-\frac{1}{m}}^2$$

$$= \left(4 - \frac{8}{3} - \frac{2}{m} + \frac{4}{m} \right) - \left(\frac{1}{m^2} + \frac{1}{3m^3} - \frac{1}{2m^3} - \frac{2}{m^2} \right)$$

$$= \frac{4}{3} + \frac{2}{m} + \frac{1}{m^2} + \frac{1}{6m^3}$$

$$(i) A = \frac{9}{2}$$

$$\frac{9}{2} = \frac{4}{3} + \frac{2}{m} + \frac{1}{m^2} + \frac{1}{6m^3}$$



$$27m^3 = 8m^3 + 12m^2 + 6m + 1$$

$$19m^3 - 12m^2 - 6m - 1 = 0$$

$$(m-1)(19m^2 + 7m + 1) = 0$$

$$D < 0$$

$$m = 1$$

(ii) minimum area

$$\frac{dA}{dm} = \frac{-2}{m^2} - \frac{2}{m^3} - \frac{1}{2m^4} = 0$$

$$\frac{4m^2 + 4m + 1}{2m^4} = 0$$

$$(2m+1)^2 = 0 \text{ or } m = -\frac{1}{2}$$

$$m = -\frac{1}{2} \text{ or } m = \frac{1}{0}$$

$$m \neq -\frac{1}{2}$$

$$\text{So } m = \frac{1}{0}$$

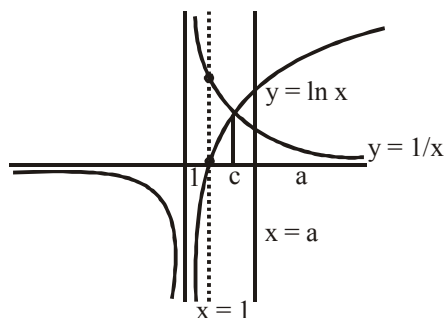
$$A_{\min} = \frac{4}{3}$$

18. Consider two curves $C_1 : y = \frac{1}{x}$ and $C_2 : y = \ln x$ on the xy plane. Let D_1 denotes the region surrounded by C_1, C_2 and the line $x = 1$ and D_2 denotes the region surrounded by C_1, C_2 and the line $x = a$. If $D_1 = D_2$. Find the value of 'a'.

$C_1 : y = \frac{1}{x}$ और $C_2 : y = \ln x$, xy समतल में दो वक्र हैं। माना D_1 वह क्षेत्र है जो कि C_1, C_2 और रेखा $x = 1$ से परिबद्ध है और D_2, C_1, C_2 और रेखा $x = a$ से परिबद्ध क्षेत्र है। यदि $D_1 = D_2$ हो तब a का मान ज्ञात कीजिए।

Ans. e

Sol.



$$D_1 = D_2$$

$$\int_1^c \left(\frac{1}{x} - \ln x \right) dx = \int_c^a \left(\ln x - \frac{1}{x} \right) dx = 0$$

$$\int_1^c \left(\frac{1}{x} - \ln x \right) dx + \int_c^a \left(\frac{1}{x} - \ln x \right) dx = 0$$

$$\int \left(\frac{1}{x} - \ln x \right) dx = 0$$

$$\ln x - x \left| x \left(\frac{x}{e} \right) \right|_1^a = 0$$

$$\ln a - a \ln \left(\frac{a}{e} \right) + \ln \left(\frac{1}{e} \right) = 0$$

$$\ln a - a \ln a + a - 1 = 0$$

$$(\ln a - 1)(a - 1) = 0$$

$$\ln a = 1 \text{ \& } a = 1$$

$$a = e$$

19. Find the value(s) of the parameter 'a' ($a > 0$) for each of which the area of the figure bounded by the straight line,

$$y = \frac{a^2 - ax}{1 + a^4} \text{ \& the parabola } y = \frac{x^2 + 2ax + 3a^2}{1 + a^4} \text{ is the greatest.}$$

∴ इसके मानों का ज्ञात कीजिए जबकि सरल रेखा $y = \frac{a^2 - ax}{1 + a^4}$ और परवलय $y = \frac{x^2 + 2ax + 3a^2}{1 + a^4}$ से परिबद्ध क्षेत्रफल अधिकतम है।

Ans. $a = 3^{1/4}$

Sol. $A = \int_{-a}^{-2a} \frac{a^2 - ax - (x^2 + 2ax + 3a^2)}{1 + a^4} dx$

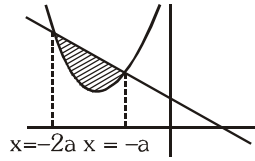
$$= \frac{3}{2} \frac{a^3}{1 + a^4}$$

Now $f(a) = \frac{3}{2} \frac{a^3}{1 + a^4}$

$$\Rightarrow f'(a) = 0$$

$$\Rightarrow (1 + a^4) 3a^2 - a^3 4a^3 = 0$$

$$\Rightarrow a_{\min} = 0, a_{\max} = 3^{1/4}$$



20. For what value of 'a' is the area bounded by the curve $y = a^2x^2 + ax + 1$ and the straight line $y = 0$, $x = 0$ and $x = 1$ the least?

के किस मान के लिए वक्र $y = a^2x^2 + ax + 1$ और सरल रेखा $y = 0$, $x = 0$ और $x = 1$ से परिबद्ध क्षेत्रफल न्यूनतम होगा।

Ans. $a = -3/4$

Sol. Required area $= \int_0^1 y dx$

$$A = \int_0^1 (a^2x^2 + ax + 1) dx$$

$$A = \frac{a^2x^3}{3} + \frac{ax^2}{2} + x \Big|_0^1$$

$$A = \frac{a^2}{3} + \frac{a}{2} + 1$$

For min

$$\frac{dA}{da} = 0$$

$$\frac{2a}{3} + \frac{1}{2} = 0$$

$$\Rightarrow a = -\frac{3}{4}$$

21. Find the positive value of 'a' for which the parabola $y = x^2 + 1$ bisects the area of the rectangle with vertices $(0, 0)$, $(a, 0)$, $(0, a^2 + 1)$ and $(a, a^2 + 1)$.

का वह धनात्मक मान ज्ञात कीजिए जिसके लिए परवलय $y = x^2 + 1$ चतुर्भुज जिनके शीर्ष $(0, 0)$, $(a, 0)$, $(0, a^2 + 1)$ और $(a, a^2 + 1)$ से परिबद्ध क्षेत्रफल को समद्विभाजित करता है।

Ans. $\sqrt{3}$

Sol. Area of rectangle = $a(a^2 + 1)$
 $= a^3 + a$

$$A = \int_0^a y dx$$

$$\frac{a^3 + a}{2} = \frac{a^3}{3}$$

$$a = 0, a = \pm\sqrt{3} \Rightarrow a = \sqrt{3}$$

22. Find the value of 'c' for which the area of the figure bounded by the curve, $y = 8x^2 - x^5$, the straight lines $x = 1$ & $x = c$ & the abscissa axis is equal to $16/3$.

ब का मान ज्ञात कीजिए जबकि वक्र $y = 8x^2 - x^5$, सरल रेखाओं $x = 1$ और $x = c$ तथा भुज अक्ष से परिबद्ध क्षेत्रफल $16/3$ वर्ग इकाई है।

Ans. $C = -1$ or $(8 - \sqrt{17})^{1/3}$

Sol. C-I $C < 1$

$$\int_c^1 (8x^2 + x^5) dx = \frac{16}{3}$$

$$\left. \frac{8x^3}{3} - \frac{x^6}{6} \right|_c^1 = \frac{16}{3}$$

$$\frac{8}{3} - \frac{1}{6} - \frac{8c^3}{3} + \frac{c^6}{6} = \frac{16}{3}$$

$$\frac{-8}{3}c^3 + \frac{c^6}{6} = \frac{17}{6} = \frac{17}{6}$$

$$c^6 - 16c^3 - 17 = 0$$

$$c^3 = -1, 17$$

$$C = -1, 17^{1/3}$$

$$C < 1$$

$$C = -1$$

$$C - II$$

$$C \geq 1$$

$$\int_1^C (8x^2 - x^5) dx = \frac{16}{3}$$

$$\left. \frac{8x^3}{3} - \frac{x^6}{6} \right|_1^C = \frac{16}{3}$$

$$\frac{8C^3}{3} - \frac{C^6}{6} = \frac{47}{6}$$

$$C^6 - 16C^3 + 47 = 0$$

$$C^3 = t$$

$$t^2 - 16t + 47 = 0$$

$$t = \frac{16 \pm \sqrt{256 - 188}}{2}$$

$$t = 8 \pm \sqrt{64 - 47}$$

$$t = 8 \pm \sqrt{17}$$

$$C^3 = 8 \pm \sqrt{17}$$

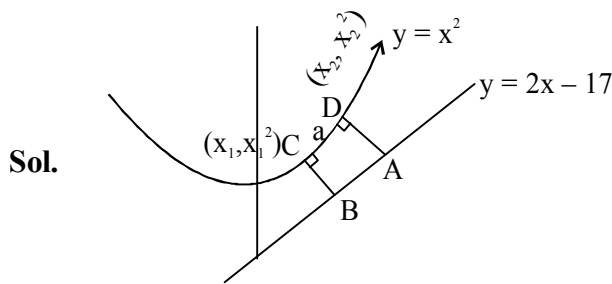
$$C = (8 + \sqrt{17})^{1/3}, (8 - \sqrt{17})^{1/3}$$

23. Consider one side AB of a square ABCD, (read in order) on the line $y = 2x - 17$, and the other two vertices C, D on the parabola $y = x^2$.
- (a) Find the minimum intercept of the line CD on y-axis.
- (b) Find the maximum possible area of the square ABCD.
- (c) Find the area enclosed by the line CD with minimum y-intercept and the parabola $y = x^2$

माना वर्ग ABCD (क्रम में) की एक भुजा AB, सरल रेखा $y = 2x - 17$ पर है तथा अन्य दो शीर्ष C, D परवलय $y = x^2$ पर

- (a) सरल रेखा CD का न्यूनतम y-अन्तःखण्ड ज्ञात कीजिए।
- (b) वर्ग ABCD का अधिकतम सम्भव क्षेत्रफल ज्ञात कीजिए।
- (c) वह क्षेत्रफल ज्ञात कीजिए जो कि रेखा CD का न्यूनतम y-अन्तःखण्ड और परवलय $y = x^2$ से परिबद्ध है

Ans. (a) 3 (b) 1280 (c) $\frac{32}{3}$



(a) As ABCD is a square $\therefore AB \parallel CD$

Equation of line CD $\equiv y = 2x + c$

$$\text{distance between AB \& CD} = \frac{|c+17|}{\sqrt{2^2+1}} = \frac{|c+17|}{\sqrt{5}}$$

be 'a'

To find x coordinates of C and D solve

$$y = x^2 \text{ and } y = 2x + c$$

$$x^2 = 2x + c$$

$$x^2 - 2x - c = 0 \begin{cases} x_1 \\ x_2 \end{cases}$$

$$\Rightarrow x_1 + x_2 = 2$$

Now length of CD

$$= \sqrt{(x_2 - x_1)^2 + (x_2^2 - x_1^2)^2}$$

$$= \sqrt{(x_2 - x_1)^2 + (x_2 - x_1)^2 (x_2 + x_1)^2}$$

$$= \sqrt{(x_2 - x_1)^2 \{1 + 4\}}$$

$$= \sqrt{5} |x_2 - x_1|$$

$$\text{Now } |x_2 - x_1| = \frac{\sqrt{D}}{|a'|}$$

$$= \sqrt{4 + 4c}$$

$$= 2\sqrt{1 + c}$$

$$\therefore a = 2\sqrt{5}\sqrt{1 + c} = \frac{|c + 17|}{\sqrt{5}}$$

$$4.5^2 (1 + c) = (c + 17)^2$$

$$100(1 + c) = c^2 + 34c + 289$$

$$100 + 100c = c^2 + 34c + 289$$

$$c^2 - 66c + 189 = 0$$

$$c = 63, 3$$

$$\therefore c_{\min} = 3$$

$$c_{\max} = 63$$

$$(b) \text{ Maximum area } = a^2$$

$$= (2\sqrt{5}\sqrt{1 + c})^2$$

$$= 4.5(1 + c)$$

$$= 20(1 + 63)$$

$$= 20 \times 64$$

$$= 1280$$

$$(c) \text{ Equation of line CD with min y intercept i.e. } c = 3$$

$$\text{i.e. } c = 3$$

$$y = 2x + 3 \text{ and } y = x^2$$

Point of intersection

$$x^2 = 2x + 3$$

$$\therefore x^2 - 2x - 3 = 0$$

$$x = 3, -1$$

$$\therefore \int_{-1}^3 ((2x + 3) - x^2) dx$$

$$-\frac{x^3}{3} + x^2 + 3x \Big|_{-1}^3$$

$$\Rightarrow \frac{32}{3}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 24 to 26

Consider the polynomial $f(x) = 1 + 2x + 3x^2 + 4x^3$. Let s be the sum of all distinct real roots of $f(x)$ and let $t = |s|$.

माना $f(x) = 1 + 2x + 3x^2 + 4x^3$ एक बहुपदीय फलन है तथा S , $f(x)$ के सभी वास्तविक भिन्न-भिन्न मूलों का योग है और माना $t = |s|$.

24. The real number s lies in the interval :

[JEE-2010]

वास्तविक संख्या s अन्तराल में होगी :

[JEE-2010]

- (A) $\left(-\frac{1}{4}, 0\right)$ (B) $\left(-11, -\frac{3}{4}\right)$ (C) $\left(-\frac{3}{4}, -\frac{1}{2}\right)$ (D) $\left(0, \frac{1}{4}\right)$

Ans. C

Sol. $\because f(x) = 2 + 6x + 12x^2 > 0 \forall x \in \mathbb{R}$

$\therefore f(x)$ is strictly increasing in $\mathbb{R}$

$$\because f(0) = 1, f(-1) = -2, f\left(-\frac{1}{2}\right) = \frac{1}{4} \text{ and } f\left(-\frac{3}{4}\right) = -\frac{1}{2}$$

$$\therefore f(x) = 0 \text{ has only one real root lying in } \left(-\frac{3}{4}, -\frac{1}{2}\right)$$

25. The area bounded by the curve $y = f(x)$ and the lines $x = 0$, $y = 0$ and $x = t$, lies in the interval :

[JEE-2010]

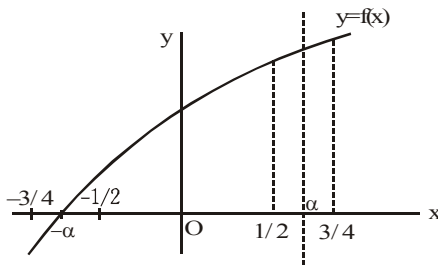
$y = f(x)$ सरल रेखाओं $x = 0$, $y = 0$ और $x = t$ से परिबद्ध क्षेत्रफल किस अन्तराल में होगा :

[JEE-2010]

- (A) $\left(\frac{3}{4}, 3\right)$ (B) $\left(\frac{21}{64}, \frac{11}{16}\right)$ (C) (9, 10) (D) $\left(0, \frac{21}{64}\right)$

Ans. A

Sol.



Let real root is $-\alpha$

$$\Rightarrow t = |s| = \alpha$$

Required area

$$A = \int_0^{\alpha} f(x)dx \quad \text{and} \quad \int_0^{1/2} f(x)dx < A < \int_0^{3/4} f(x)dx$$

$$\Rightarrow \left| x + x^2 + x^3 + x^4 \right|_0^{1/2} < A < \left| x + x^2 + x^3 + x^4 \right|_0^{3/4} < \left| 4x \right|_0^{3/4}$$

$$\Rightarrow \frac{15}{16} < A < 3$$

26. The function $f'(x)$ is :

[JEE-2010]

(A) increasing in $\left(-t, -\frac{1}{4}\right)$ and decreasing in $\left(-\frac{1}{4}, t\right)$

(B) decreasing in $\left(-t, -\frac{1}{4}\right)$ and increasing in $\left(-\frac{1}{4}, t\right)$

(C) increasing in $(-t, t)$

(D) decreasing in $(-t, t)$

फलन $f'(x)$ है :

[JEE-2010]

(A) $\left(-t, -\frac{1}{4}\right)$ में वर्धमान और $\left(-\frac{1}{4}, t\right)$ में ह्यसमान

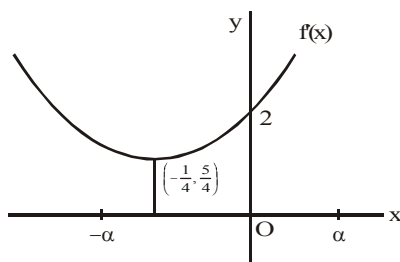
(B) $\left(-t, -\frac{1}{4}\right)$ में ह्यसमान और $\left(-\frac{1}{4}, t\right)$ में वर्धमान

(C) ; $\forall x$ में वर्धमान

(D) ; $\forall x$ में ह्यसमान

Ans. B

Sol.



$$f(x) = 2(6x^2 + 3x + 1)$$

$$\Rightarrow f(x) \text{ is decreasing in } \left(-\alpha, -\frac{1}{4}\right) \text{ increasing in } \left(-\frac{1}{4}, \alpha\right)$$

$$\text{or } f(x) \text{ is decreasing in } \left(-t, -\frac{1}{4}\right) \text{ and increasing in } \left(-\frac{1}{4}, t\right)$$

Paragraph for Question nos. 27 to 29

Consider three real-valued functions f, g and h defined on $\mathbb{R}$ (the set of real numbers). Let $f(x) = 2x^3 + 3\left(1 - \frac{3a}{2}\right)x^2 + 3(a^2 - a)x + b$ where $a, b \in \mathbb{R}$ and $g(x) = \frac{f'(x)}{6}$. Also $h(x)$ is such that $h''(x) = 6x - 4$ and $h(x)$ has a local minimum value 5 at $x = 1$.

वास्तविक मान वाले फलनों f, g तथा h जो कि $\mathbb{R}$ (वास्तविक संख्याओं के समुच्चय) पर परिभाषित है, माना $f(x) = 2x^3 + 3\left(1 - \frac{3a}{2}\right)x^2 + 3(a^2 - a)x + b$ जहाँ $a, b \in \mathbb{R}$ तथा $g(x) = \frac{f'(x)}{6}$. तथा $h(x)$ इस प्रकार है कि $h''(x) = 6x - 4$ तथा $h(x)$ का, $x = 1$ पर स्थानीय निम्निष्ठ 5 है।

27. The true set of values of a for which $f(x)$ has negative point of local minimum, is :

a का वास्तविक समुच्चय जिनके लिये $f(x)$ के पास स्थानीय निम्निष्ठ का ऋणात्मक बिन्दु है :

- (A) $(-\infty, 0)$ (B) $(1, \infty)$ (C) $(0, 1)$ (D) $(1, \infty) - \{2\}$

Ans. A

Sol. We have

$$f(x) = 2x^3 + 3\left(1 - \frac{3a}{2}\right)x^2 + 3(a^2 - a)x + b$$

$$\text{So, } f'(x) = 6x^2 + 6\left(1 - \frac{3a}{2}\right)x + 3(a^2 - a)$$

Now, for $f(x)$ to have negative point of local minimum, we must have both roots of the equation $f'(x) = 0$ real (unequal) and negative.

$$\therefore \text{Discriminant} > 0 \Rightarrow 36 \left[\left(1 - \frac{3a}{2}\right)^2 - 2(a^2 - a) \right] > 0 \Rightarrow \left(\frac{a}{2} - 1\right)^2 > 0 \Rightarrow a \in \mathbb{R} - \{2\} \quad \dots\dots\dots(1)$$

$$\text{Also, sum of roots} < 0 \Rightarrow \frac{-6\left(1 - \frac{3a}{2}\right)}{6} < 0 \Rightarrow \frac{3a}{2} - 1 < 0 \Rightarrow a < \frac{2}{3} \quad \dots\dots\dots(2)$$

$$\text{And product of roots} > 0 \Rightarrow \frac{3(a^2 - a)}{6} > 0 \Rightarrow a(a - 1) > 0 \Rightarrow a \in (-\infty, 0) \cup (1, \infty) \quad \dots\dots\dots(3)$$

$\therefore$ From $(1) \cap (2) \cap (3)$, we get $a \in (-\infty, 0)$

28. The complete set of values of a for which vertex of parabola $y = g(x)$ has negative ordinate, is :

a के वास्तविक मानों का समुच्चय जिनके लिये, परवलय $y = g(x)$ का शीर्ष ऋणात्मक कोटि का है :

- (A) $\left(\frac{1}{2}, 1\right)$ (B) $(0, \infty)$ (C) $\mathbb{R} - \{2\}$ (D) $(1, \infty) - \{2\}$

Ans. C

Sol. We have

$$g(x) = \frac{f'(x)}{6} = x^2 + \left(1 - \frac{3a}{2}\right)x + \left(\frac{a^2 - a}{2}\right)$$

As discriminant of $g(x) = \left(\frac{a}{2} - 1\right)^2 > 0 \forall a \in \mathbb{R} - \{2\}$

$\Rightarrow$ Ordinate of the vertex of parabola $y = g(x)$ lies below x-axis.

So, $a \in \mathbb{R} - \{2\}$

29. The area bounded by $y = h(x)$ between $x = 0$ and $x = 2$, is :

$y = h(x)$, से $x = 0$ एवं $x = 2$ के मध्य का क्षेत्रफल है :

- (A) $\frac{23}{3}$ (B) $\frac{20}{3}$ (C) $\frac{40}{3}$ (D) $\frac{32}{3}$

Ans. D

Sol. We have

$$h''(x) = 6x - 4 \Rightarrow h'(x) = 3x^2 - 4x + c$$

$$\text{As } h'(1) = 0, \text{ so } 0 = 3 - 4 + c \Rightarrow c = 1$$

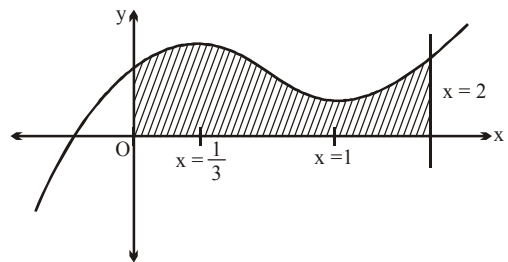
$$\therefore h'(x) = 3x^2 - 4x + 1 \Rightarrow h(x) = x^3 - 2x^2 + x + k$$

$$\text{As } h(1) = 5, \text{ so } 5 = 1 - 2 + 1 + k \Rightarrow k = 5.$$

$$\text{Hence, } h(x) = x^3 - 2x^2 + x + 5$$

Clearly, required area

$$= \int_0^2 h(x) dx = \int_0^2 (x^3 - 2x^2 + x + 5) dx = \left(\frac{x^4}{4} - \frac{2x^3}{3} + \frac{x^2}{2} + 5x \right)_0^2 = \frac{32}{3} \text{ (square units).}$$



Paragraph for Question Nos. 30 to 32

Consider the functions defined implicitly by the equation $y^3 - 3y + x = 0$ on various intervals in the real line. If $x \in (-\infty, -2) \cup (2, \infty)$, the equation implicitly defines a unique real valued differentiable function $y = f(x)$. If $x \in (-2, 2)$, the equation implicitly defines a unique real valued differentiable function $y = g(x)$ satisfying $g(0) = 0$.

माना एक फलन अस्पष्ट रूप में समीकरण $y^3 - 3y + x = 0$ द्वारा वास्तविक संख्या रेखा के अन्तरालों के लिये परिभाषित है यदि $x \in (-\infty, -2) \cup (2, \infty)$ तब समीकरण एक अद्वितीय वास्तविक मान वाला अवकलनीय फलन $y = f(x)$ द्वारा परिभाषित है और यदि $x \in (-2, 2)$ तब समीकरण, अद्वितीय वास्तविक अवकलनीय फलन $y = g(x)$, $g(0) = 0$, द्वारा परिभाषित है।

30. If $f(-10\sqrt{2}) = 2\sqrt{2}$, then $f''(-10\sqrt{2}) =$

यदि $f(-10\sqrt{2}) = 2\sqrt{2}$, तो $f''(-10\sqrt{2}) =$

(A) $\frac{4\sqrt{2}}{7^3 3^2}$ (B) $-\frac{4\sqrt{2}}{7^3 3^2}$ (C) $\frac{4\sqrt{2}}{7^3 3}$ (D) $-\frac{4\sqrt{2}}{7^3 3}$

Ans. B

Sol. $y^3 - 3y + x = 0$

$$3y^2 y' - 3y' + 1 = 0 \quad y' = \frac{-1}{3(y^2 - 1)}$$

$$f(-10\sqrt{2}) = 2\sqrt{2}$$

$$f'(-10\sqrt{2}) = -\frac{1}{3(7)} = -\frac{1}{21}$$

$$6y(y')^2 + 3y^2 y'' - 3y'' = 0$$

$$y'' = -\frac{2y(y')^2}{y^2 - 1}$$

$$f''(-10\sqrt{2}) = \frac{-2(2\sqrt{2})}{441 \times 7} = \frac{-4\sqrt{2}}{7^3 3^2}$$

31. The area of the region bounded by the curve $y = f(x)$, the x-axis, and the lines $x = a$ and $x = b$, where $-\infty < a < b < -2$, is :

चक्र $y = f(x)$, x-अक्ष और रेखाओं $x = a$ और $x = b$ जहाँ $-\infty < a < b < -2$, द्वारा परिबद्ध क्षेत्रफल है :

(A) $\int_a^b \frac{x}{3[(f(x))^2 - 1]} dx + bf(b) - af(a)$ (B) $-\int_a^b \frac{x}{3[(f(x))^2 - 1]} dx + bf(b) - af(a)$

(C) $\int_a^b \frac{x}{3[(f(x))^2 - 1]} dx - bf(b) + af(a)$ (D) $-\int_a^b \frac{x}{3[(f(x))^2 - 1]} dx - bf(b) + af(a)$

Ans. A

Sol. $\int_a^b f(x) dx = [xf(x)]_{ab} - \int_a^b xf'(x) dx$

$$= bf(b) - af(a) + \int_a^b \frac{x}{3[(f(x))^2 - 1]} dx$$

$$= \int_a^b \frac{x}{3[(f(x))^2 - 1]} dx + bf(b) - af(a)$$

32. $\int_{-1}^1 g'(x) dx =$

- (A) $2g(-1)$ (B) 0 (C) $-2g(1)$ (D) $2g(1)$

Ans. D

Sol. $y^3 - 3y + x = 0$

when $x \in (-2, 2)$ $y = g(x)$ and $g(0) = 0$

$$(g(x))^3 - 3g(x) + x = 0$$

$$(g(-x))^3 - 3g(-x) - x = 0$$

$$+ \quad + \quad +$$

$$(g(x))^3 + (g(-x))^3 - 3(g(x) + g(-x)) = 0$$

$$\{g(x) + g(-x)\}^2 \{(g(x))^2 + (g(-x))^2 - g(x) \cdot g(-x) - 3\} = 0$$

$$\Rightarrow g(x) + g(-x) = 0$$

$$\Rightarrow g(x) = -g(-x) \quad \text{_____ (A)}$$

$\therefore g(x)$ is odd

$$\text{Now } \int_{-1}^1 g'(x) dx = g(x) \Big|_{-1}^1$$

$$= g(1) - g(-1) \quad \text{_____ (1)}$$

Put $x = 1$ in A

$$g(1) = -g(-1)$$

$$g(-1) = -g(1)$$

$$\therefore \text{Equation becomes } g(1) - (-g(1))$$

$$\Rightarrow 2g(1)$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

33. Area of the region bounded by the curve $y = e^x$ and lines $x = 0$ and $y = e$ is :

[JEE-2009]

वक्र $y = e^x$ और रेखाओं $x = 0$ और $y = e$ द्वारा परिबद्ध क्षेत्रफल है :

[JEE-2009]

- (A) $e - 1$ (B) $\int_1^e \ln(e+1-y)dy$ (C) $e - \int_0^1 e^x dx$ (D) $\int_1^e \ln y dy$

Ans. BCD

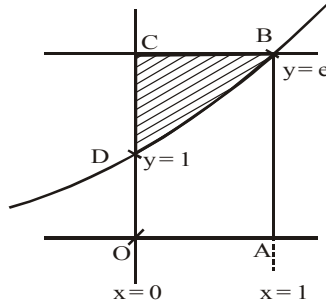
Sol. $A = \int_1^e \ln y dy$

Apply

$$= \int_1^e \ln(e+1-y)dy$$

$$A = \text{ar}(\text{OABC}) - \text{ar}(\text{OABD})$$

$$= e - \int_0^1 e^x dx$$

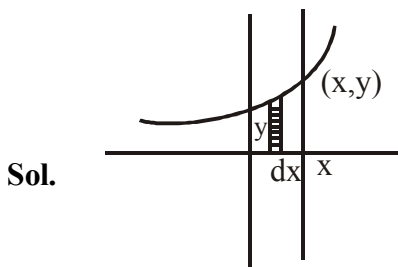


34. The area bounded by a curve, the axis of co-ordinates & the ordinate of some point of the curve is equal to the length of the corresponding arc of the curve . If the curve passes through the point $P(0, 1)$ then the equation of this curve can be :

वक्र, निर्देशी अक्षों तथा वक्र के किसी बिन्दु की कोटि द्वारा घिरा क्षेत्रफल, वक्र की संगत चाप की लम्बाई के बराबर है तथा वक्र बिन्दु $P(0, 1)$ से गुजरता है तब वक्र का समीकरण होगा :

- (A) $y = \frac{1}{2}(e^x - e^{-x} + 2)$ (B) $y = \frac{1}{2}(e^x + e^{-x})$ (C) $y = 1$ (D) $y = \frac{2}{e^x + e^{-x}}$

Ans. BC



Sol.

$$A = \int_0^x y dx = \int_0^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$y^2 = 1 + \left(\frac{dy}{dx}\right)^2$$

Case-I $y \neq 1$

Case-II $y = 1$

$$\int \frac{dy}{\sqrt{y^2 - 1}} = \pm \int dx \quad \because y \neq 1$$

$$\ln |y + \sqrt{y^2 - 1}| = x + C$$

$$(0, 1) \Rightarrow C = 0$$

$$\ln |y + \sqrt{y^2 - 1}| = x$$

$$y + \sqrt{y^2 - 1} > 0$$

$$y + \sqrt{y^2 - 1} = e^x$$

$$y - \sqrt{y^2 - 1} = e^{-x}$$

$$\text{add } y = \frac{e^x + e^{-x}}{2}$$

35. Let T be the triangle with vertices (0, 0), (0, c^2) and (c , c^2) and let R be the region between $y = cx$ and $y = x^2$ where $c > 0$ then :

यदि T एक त्रिभुज है जिसके शीर्ष (0, 0), (0, c^2) तथा (c , c^2) है तथा यदि R, $y = cx$ एवं $y = x^2$ द्वारा घिरा क्षेत्र है तो : (जहाँ $c > 0$)

$$(A) \text{ Area (R)} = \frac{c^3}{6}$$

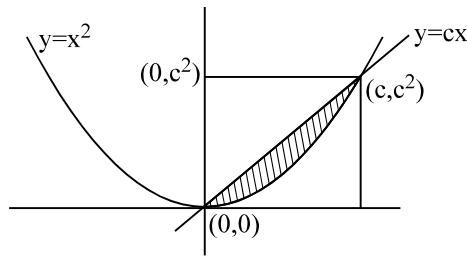
$$(B) \text{ Area of R} = \frac{c^3}{3}$$

$$(C) \lim_{c \rightarrow 0^+} \frac{\text{Area (T)}}{\text{Area (R)}} = 3$$

$$(D) \lim_{c \rightarrow 0^+} \frac{\text{Area (T)}}{\text{Area (R)}} = \frac{3}{2}$$

Ans. AC

Sol. $\text{Area (T)} = \frac{c \cdot c^2}{2} = \frac{c^3}{2}$



$$\begin{aligned} \text{Area (R)} &= \frac{c^3}{2} - \int_0^c x^2 dx \\ &= \frac{c^3}{2} - \frac{c^3}{3} = \frac{c^3}{6} \end{aligned}$$

$$\therefore \lim_{c \rightarrow 0^+} \frac{\text{Area (T)}}{\text{Area (R)}} = \lim_{c \rightarrow 0^+} \frac{c^3}{2} \cdot \frac{6}{c^3} = 3$$

36. Suppose f is defined from $\mathbb{R} \rightarrow [-1, 1]$ as $f(x) = \frac{x^2 - 1}{x^2 + 1}$ where $\mathbb{R}$ is the set of real number. Then the statement which does not hold is :
- (A) f is many one onto
 (B) f increases for $x > 0$ and decrease for $x < 0$
 (C) minimum value is not attained even though f is bounded
 (D) the area included by the curve $y = f(x)$ and the line $y = 1$ is π sq. units

माना $f, \mathbb{R} \rightarrow [-1, 1]$ में परिभाषित है $f(x) = \frac{x^2 - 1}{x^2 + 1}$ जहाँ $\mathbb{R}$ वास्तविक संख्याएं है तब कौनसा कथन असत्य है :

- (A) f बहुएकैकी आच्छादक है
 (B) $f, x > 0$ में वर्धमान तथा $x < 0$ में ह्रासमान
 (C) यदि f परिवर्द्ध है तो भी न्यूनतम मान गृहण नहीं करता
 (D) वक्र $y = f(x)$ तथा रेखा $y = 1$ द्वारा परिबद्ध क्षेत्रफल π वर्ग ईकाई है

Ans. ACD

Sol. $y = f(x) = \frac{x^2 - 1}{x^2 + 1} = 1 - \frac{2}{x^2 + 1}$

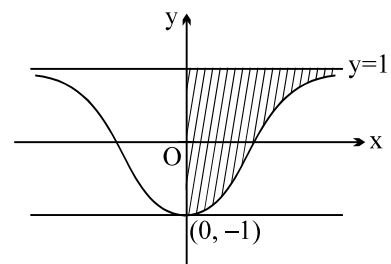
$$f'(x) = \frac{4x}{(x^2 + 1)^2}$$

$x > 0$, f is increasing and $x < 0$ f is decreasing $\Rightarrow$ (B) is true

range is $[-1, 1) \Rightarrow$ into $\Rightarrow$ (A) is false;

minimum value occurs at $x = 0$ and $f(0) = -1 \Rightarrow$ (C) is false

$$A = 2 \int_0^\infty \left(1 - \frac{x^2 - 1}{x^2 + 1}\right) dx = 4 \int_0^\infty \frac{dx}{x^2 + 1} = 4 \cdot \tan^{-1} \Big|_0^\infty = 4 \cdot \frac{\pi}{2} = 2\pi \Rightarrow \text{(D) is false]}$$



MATCH THE COLUMN TYPE QUESTION(S)

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37. Match the following column :

Column I

(A) The area between the curves $y = 2x^4 - x^2$, the x -axis and the

Column II

(P) $2 \sin 1$